

Gold-Free Faithfulness Metrics for NL-to-FOL Translation: Internal Research Report

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1 Framing

Translating natural language into first-order logic (NL-FOL) is a prerequisite for neurosymbolic reasoning pipelines, yet evaluating whether a candidate formula faithfully captures a sentence’s meaning remains an open problem. Gold FOL annotations are expensive, non-unique, and unreliable: Brunello et al. found that approximately 42% of entries in both FOLIO and MALLS contain incorrect formalisations. Existing metrics either require a gold formula (exact match, prover-checked equivalence) or only verify that the output parses. The user’s own pilot study used structural consistency checks (arity agreement, predicate-set overlap across reruns, joint satisfiability) that measure stability but not whether a formula says what its sentence says.

This run investigates gold-free metrics that predict faithfulness to the text given only a sentence and a candidate formula, with no reference formula, no ontology, and no domain knowledge. The evaluation is a meta-evaluation: we build a labelled set of (sentence, formula) pairs from public benchmarks whose gold has been audited, and measure each metric’s ability to discriminate correct translations from errors.

2 Iteration 1

2.1 Strategy

The iteration screens four candidate metric families head-to-head on a frozen evaluation set, with a disguised cheap LLM judge as the bar to beat.

Candidates. (A) FOL-Triage, a three-layer cascade: L1 (formula lint: deterministic formula smells), L2 (text accounting: bag-of-words and role-aware coverage), and L3 (role questionnaire: a small LLM reads the text and z3 reads the formula; their structured answers are compared). (B) Round-trip verification: verbalise the formula back to natural language and compare to the original via NLI or embedding similarity. (C) Cross-system consensus: collect translations of the same sentence from multiple independent systems, measure z3-equivalence between the candidate and the peer pool, and score disagreement. (D) Disguised cheap judge and baselines: an LLM judge, structural pilot metrics, and round-trip features evaluated on nonce-disguised sentences to control for memorisation.

Evaluation set. The frozen screen uses Logic-LM released outputs on FOLIO validation: three LLM systems (GPT-3.5-turbo, GPT-4, text-davinci-003) producing 796 items. Track L contains LLM outputs labelled by z3 equivalence to corrected gold; track H contains the 294 original human gold formulas, of which 57 are errors according to the corrected labels.

Decision rule. A candidate must achieve AUROC ≥ 0.65 on track L (ERROR vs CORRECT), with coverage $\geq 90\%$, rewrite false-alarm rate $\leq 10\%$ per rewrite family, and cost $\leq \$0.002$ per item.

Development dataset (Artifact E; `gen_art_dataset.1`). A separate 700-sentence, 8,507-row development set was constructed from FOLIO training premises, MALLS training long/conditional/exception strata, and FOLIO curated conclusions, each translated by 9 LLM families plus ccg2lambda plus GPT-4 gold. Labels were assigned by z3 equivalence to audited gold, with contested cases adjudicated by a three-member panel (Claude Haiku-4.5, GLM-4.6, Kimi-K2). This set is reserved for iteration 2 confirmation. The dataset contains 1,750 CORRECT, 5,005 ERROR, 1,086 UNPARSEABLE, 276 UNRESOLVED, and 390 CONTESTED items. The dataset cost \$9.83 in API calls.

Table 1: Dataset E composition by stratum. `gen_art_dataset_1`.

Stratum	Sentences	LLM rows	CORRECT (A+B)	ERROR (A+B)	Testable	CONTESTED
L25 (all 300)	300	2960	176	697	YES	130
L20	150	1850	229	592	YES	91
EXC	100	1240	222	384	YES	90
CTRL	150	1850	237	149	YES	65

Table 2: Gold audit: reference error rates by stratum. `gen_art_dataset_1`.

Stratum	Audited	Judged wrong	Gold-error rate [Wilson 95%]
L25 (all 300)	294	249	0.847 [0.801, 0.884]
L20	146	118	0.808 [0.737, 0.864]
EXC	95	73	0.768 [0.674, 0.842]
CTRL	148	112	0.757 [0.682, 0.819]
MALLS all	535	440	0.822 [0.788, 0.853]

Table 3: Panel calibration: synthetic gate (prompt v2) and real-error accuracy. `gen_art_dataset_1`.

Member	Model	n	Bal. acc	Rec F	Rec U	Gate
P1	Haiku-4.5	77	0.861	0.75	0.973	PASS
P3	GLM-4.6	77	0.850	0.70	1.000	PASS
R1	Kimi-K2	77	0.863	0.725	1.000	PASS

Table 4: Panel real-error accuracy (track H, 75 unambiguous pairs). `gen_art_dataset_1`.

Member	Judgements	Accuracy	Orig. flagged	Corrected accepted
P1	150	0.747	0.867	0.627
P3	150	0.693	0.773	0.613
R1	142	0.725	0.720	0.653
Majority	150	0.727	0.840	0.613

Table 5: Inter-rater agreement (P3 and R1, Cohen κ). `gen_art_dataset_1`.

Stratum	Items	Raw agreement	Cohen κ	P3 faithful rate
L25	1985	0.845	0.472	0.199
L20	1150	0.839	0.520	0.245
EXC	688	0.789	0.442	0.279
CTRL	721	0.839	0.618	0.314
ALL	4544	0.834	0.513	0.241

2.2 Experiment A: FOL-Triage (gen_art_experiment_1)

FOL-Triage was evaluated on the frozen screen with Gemini 2.5 Flash Lite as the primary LLM for L3. Total API spend was \$0.29.

Table 6: FOL-Triage headline AUROCs (track L, ERROR vs CORRECT, $n = 477$, 199 errors). Iteration 1, Experiment A.

Metric	AUROC	95% CI	AUPRC
L1 (any smell)	0.508	[0.499, 0.519]	0.425
L2 bag-of-words	0.737	[0.677, 0.801]	0.690
L2 role-aware	0.557	[0.527, 0.594]	0.476
L3 score	0.756	[0.701, 0.809]	0.657
Fused (fitted on H, no leak)	0.759	[0.695, 0.825]	0.740
Fused (all H)	0.827	[0.778, 0.874]	0.771
Fused (cross-fit L)	0.864	[0.824, 0.902]	0.789
Decomposed judge (LLM reads formula)	0.716	[0.645, 0.789]	0.717
L3 with local Qwen2.5-1.5B	0.708	[0.637, 0.774]	0.603

The primary no-leak fused score is 0.759. The fused-all-H and cross-fit-L numbers (0.827, 0.864) are shown for completeness but are not valid held-out estimates.

Track H (human gold errors, $n = 258$, 57 errors). Fused out-of-fold AUROC is 0.777 [0.698, 0.847]. L1 alone reaches 0.655, compared to 0.508 on track L.

2.2.1 L1 lint: a negative result on LLM outputs

L1 flags formulas with deterministic smells (existential implication, free variables, arity clashes, vacuity, trivial formulas, dangling predicates). On track L, the true positive rate is 2.0% at a false positive rate of 0.4%, giving AUROC 0.508. This is a negative result: L1 lint does not transfer from human annotation errors (where it is effective, AUROC 0.655) to LLM outputs. LLM-generated formulas rarely contain the syntactic pathologies L1 detects; their errors are semantic.

Table 7: L1 smell false-alarm rates per system (track L). Iteration 1, Experiment A.

Smell	gpt-3.5 FA	gpt-4 FA	davinci-003 FA
EX_IMP	0.0	0.0	0.0
ALL_AND	0.0	0.0	0.0
IFF_RESTR	0.0	0.009	0.0
GLUE	0.0	0.0	0.0
FREE	0.0	0.0	0.0
VACUOUS	0.0	0.0	0.0
TRIVIAL	0.0	0.0	0.0
ARITY	0.0	0.0	0.0
DANGLING	0.0	0.0	0.0

2.2.2 L2 role-aware check: below bag-of-words

The role-aware UD-based check (L2-role) achieves AUROC 0.557, which is 0.18 below the simpler bag-of-words check (L2-bow, 0.737). The role-aware layer adds no discriminative power over word overlap.

2.2.3 Per-type sensitivity (track L, at binary threshold)

Table 8: Per-error-type sensitivity on track L. Iteration 1, Experiment A.

Error type	n	L1	L2-bow	L2-role	L3	Fused	Cascade
NEG	5	0.0	0.2	1.0	0.2	0.4	1.0
QUANT	2	0.0	0.0	0.5	0.0	0.5	0.5
CONN	4	0.0	0.0	0.0	0.0	0.0	0.0
DROP	142	0.028	0.479	0.127	0.197	0.599	0.606
ADD	150	0.020	0.473	0.107	0.173	0.653	0.573
SWAP	12	0.0	0.0	0.083	0.25	0.083	0.333
BIND	8	0.125	0.0	0.125	0.0	0.375	0.25
COMPOUND	187	0.118	0.428	0.166	0.160	0.599	0.578

FOL-Triage is most sensitive to ADD and DROP errors, with the fused score catching about 60%. It misses CONN errors entirely (0/4) and detects SWAP poorly (1/12).

Table 9: Per-error-type sensitivity on track H. Iteration 1, Experiment A.

Error type	n	L1	L2-bow	L2-role	L3	Fused	Cascade
NEG	4	0.0	0.0	0.25	0.0	0.25	0.25
REV	1	0.0	0.0	0.0	0.0	0.0	0.0
QUANT	1	0.0	0.0	0.0	0.0	0.0	0.0
RESTR	1	1.0	1.0	0.0	0.0	1.0	1.0
CONN	8	0.25	0.125	0.0	0.375	0.375	0.375
MOVE	2	0.5	0.0	0.5	0.0	0.5	1.0
DROP	14	0.143	0.357	0.0	0.143	0.357	0.5
ADD	19	0.158	0.368	0.053	0.211	0.526	0.579
SWAP	2	0.0	0.5	0.0	0.5	0.5	0.5
BIND	1	1.0	1.0	0.0	0.0	1.0	1.0
UNGLUE	5	1.0	0.4	0.0	0.2	1.0	1.0
COMPOUND	33	0.303	0.182	0.242	0.182	0.545	0.636

2.2.4 Invariance under meaning-preserving rewrites

Five rewrite families were tested on 150 track-L CORRECT items. The rewrite false-alarm gate ($\leq 10\%$ per family) fails. The worst-family fused FA is 0.275 (CONTRAPOSITIVE), but this equals the FA on the original formulas; the flip rate is at most 0.061. The high FA is not caused by the rewrites but by the fused score’s calibration (see Figure 1).

Table 10: Rewrite invariance: false-alarm rates and flip rates (track L, 150 CORRECT items). Iteration 1, Experiment A.

Rewrite	n	L1 FA	L1 fl.	bow FA	bow fl.	role FA	role fl.	L3 FA	L3 fl.	Fused FA
RENAME	150	0.0	0.0	0.18	0.093	0.013	0.0	0.033	0.007	0.22
REORDER	34	0.0	0.0	0.029	0.0	0.059	0.0	0.059	0.0	0.235
DEMORGAN	33	0.0	0.0	0.030	0.0	0.061	0.0	0.030	0.030	0.182
CONTRA.	40	0.025	0.025	0.10	0.0	0.025	0.025	0.025	0.0	0.275
PRENEX	1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

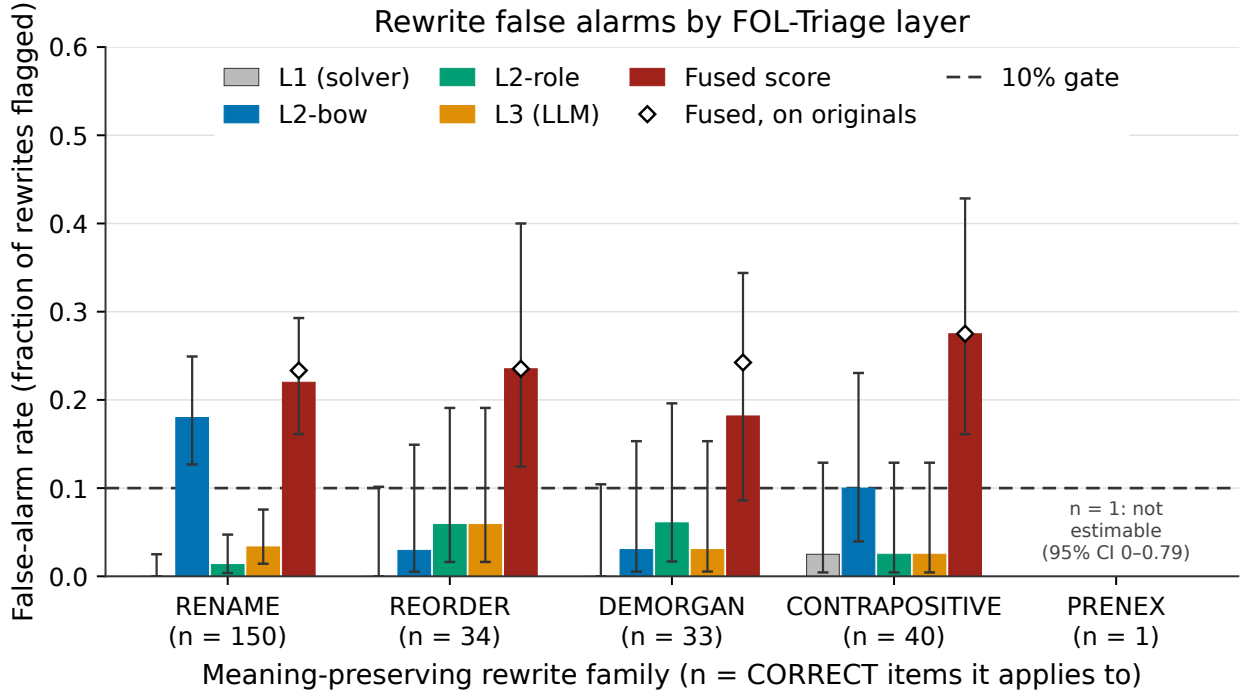


Figure 1: False-alarm rates of FOL-Triage layers and the fused score under five families of meaning-preserving rewrites (track L, CORRECT items). The fused FA is driven by calibration mismatch, not by the rewrites themselves (flip rates are near zero).

2.2.5 Coverage and cost

Table 11: Coverage rates. Iteration 1, Experiment A.

Track	n	Parseable	Full coverage rate
L	753	693	0.920
H	302	279	0.924

L1 and L2 are free (CPU-only). L3 costs \$0.175 per 1,000 calls with Gemini 2.5 Flash Lite. Total per-item cost is \$0.0002, passing the \$0.002 gate. CPU time (L1+L2+z3): median 35 ms, P95 90 ms. L3 latency: median 630 ms, P95 1.68 s.

2.2.6 Complexity strata (track L, fused score)

Table 12: Complexity strata AUROC’s (track L, fused score). Iteration 1, Experiment A.

Stratum	n (err/cor)	Fused AUROC	L2-bow	L3
words < 12	443 (179/264)	0.781	0.742	0.772
words 12–19	31 (18/13)	0.521	0.596	0.541
words ≥ 20	3 (2/1)	0.500	1.000	0.000
0 quantifiers	302 (130/172)	0.743	0.728	0.771
1 quantifier	164 (65/99)	0.794	0.748	0.721
≥ 2 quantifiers	11 (4/7)	0.750	0.768	0.607
depth ≤ 1	296 (124/172)	0.762	0.745	0.779
depth 2–3	173 (73/100)	0.756	0.717	0.710
depth ≥ 4	8 (2/6)	0.667	0.750	0.500
0 conditions	344 (145/199)	0.775	0.759	0.791
1–2 conditions	131 (54/77)	0.716	0.677	0.628

The frozen screen is dominated by short, simple sentences (93% under 12 words). The long and complex strata have too few items to test.

2.2.7 Secondary label-set AUROC’s

Table 13: Secondary label set: ERRORUCOMPOUND vs CORRECT ($n = 664$, 386 positives). `gen.art.experiment.1`.

Score	AUROC	95% CI	AUPRC
l1_any	0.532	[0.518, 0.547]	0.607
l2_bow	0.756	[0.708, 0.802]	0.806
l2_role	0.565	[0.542, 0.590]	0.634
l3_score	0.761	[0.713, 0.807]	0.776
p_fused_H	0.778	[0.723, 0.827]	0.845
p_fused_Hall	0.836	[0.793, 0.872]	0.873
cascade_llmformula	0.744	[0.687, 0.799]	0.830
l3_score_local	0.723	[0.664, 0.777]	0.750

Table 14: Secondary label set: ERROR minus pure ADD+DROP vs CORRECT ($n = 360$, 82 positives). `gen_art_experiment_1`.

Score	AUROC	95% CI	AUPRC
l1_any	0.504	[0.495, 0.519]	0.231
l2_bow	0.701	[0.614, 0.781]	0.457
l2_role	0.558	[0.523, 0.603]	0.293
l3_score	0.776	[0.701, 0.848]	0.482
p_fused_H	0.731	[0.622, 0.831]	0.496
p_fused_Hall	0.813	[0.743, 0.874]	0.546
p_fused_Lcf	0.873	[0.819, 0.916]	0.584
cascade_llmformula	0.660	[0.538, 0.777]	0.459
l3_score_local	0.731	[0.654, 0.803]	0.409

Table 15: Track H headline AUROCs ($n = 258$, 57 errors). `gen_art_experiment_1`.

Score	AUROC	95% CI	AUPRC
l1_any	0.655	[0.598, 0.716]	0.450
l2_bow	0.721	[0.646, 0.791]	0.430
l2_role	0.520	[0.480, 0.566]	0.232
l3_score	0.704	[0.626, 0.778]	0.401
p_fused_H_oof	0.777	[0.698, 0.847]	0.636

2.2.8 Binary layers at operating thresholds

Table 16: Binary layers at operating thresholds (primary set). `gen_art_experiment_1`.

Layer	TPR	FPR	Precision	Prec@10%	Prec@25%	Acc(eq)
l1_any	0.020	0.004	0.800	0.383	0.651	0.513
l2_bow_flag	0.407	0.065	0.818	0.411	0.677	0.513
l2_role_flag	0.131	0.018	0.839	0.447	0.708	0.513
l3_flag	0.176	0.018	0.875	0.521	0.765	0.534
fused_flag	0.568	0.194	0.677	0.245	0.493	0.543
l1_or_l2	0.518	0.083	0.818	0.410	0.676	0.605
cascade_any	0.553	0.097	0.803	0.387	0.655	0.513

2.2.9 L3 field gate on HREF (known-correct gold)

Gated fields: mm_role, mm_claims, mm_order, missing_frac, extra_frac. Excluded: mm_force, mm_exception. Alignment rate: 0.906.

2.2.10 Per-system AUROC (descriptive)

2.3 Experiment C: Cross-System Consensus (`gen_art_experiment_3`)

The consensus metric collects translations of the same sentence from multiple independent LLM families, checks pairwise z3 equivalence, and scores each candidate by its disagreement with the pool. Total API spend was \$1.52.

Table 17: L3 field gate accuracy on HREF (known-correct gold formulas). `gen_art_experiment_1`.

Field	Accuracy	n	Wilson 95%
mm_role	0.913	655	[0.889, 0.932]
mm_force	0.690	799	[0.657, 0.721]
mm_claims	0.863	182	[0.805, 0.905]
mm_order	0.947	57	[0.856, 0.982]
mm_exception	0.000	1	[0.000, 0.793]
missing_frac	0.916	814	[0.895, 0.934]
extra_frac	0.919	765	[0.897, 0.936]

Table 18: Per-system AUROCs (track L, 3 systems, descriptive). Iteration 1, Experiment A.

System	n	Err. rate	Fused	L1	L2-bow
gpt-3.5-turbo	143	0.378	0.689	0.500	0.650
gpt-4	177	0.367	0.700	0.496	0.716
text-davinci-003	157	0.510	0.857	0.525	0.811

Table 19: Consensus-metric AUROCs (track L, ERROR vs CORRECT, $n = 546$, 249 errors). Iteration 1, Experiment C.

Metric	AUROC	95% CI	AUPRC
c_score (all peers)	0.866	[0.821, 0.906]	0.827
c_score (6 fresh peers)	0.872	[0.830, 0.911]	0.825
c_score (non-OpenAI)	0.863	[0.820, 0.903]	0.808
c_score (2 Logic-LM peers)	0.753	[0.701, 0.806]	0.659
medoid depth	0.722	[0.673, 0.774]	0.622
cluster entropy	0.773	[0.724, 0.820]	0.691
SC-5 cheap (gpt-4.1-nano)	0.725	[0.669, 0.781]	0.627
SC-5 same model	0.716	[0.632, 0.796]	0.628
SC-5 entropy	0.631	[0.568, 0.693]	0.539
Pred-set instability (SC)	0.688	[0.626, 0.750]	0.602
Pred-set instability (pool)	0.770	[0.717, 0.821]	0.721
FOL length	0.527	[0.470, 0.586]	0.485
Misalignment (vocab)	0.835	[0.794, 0.876]	0.821

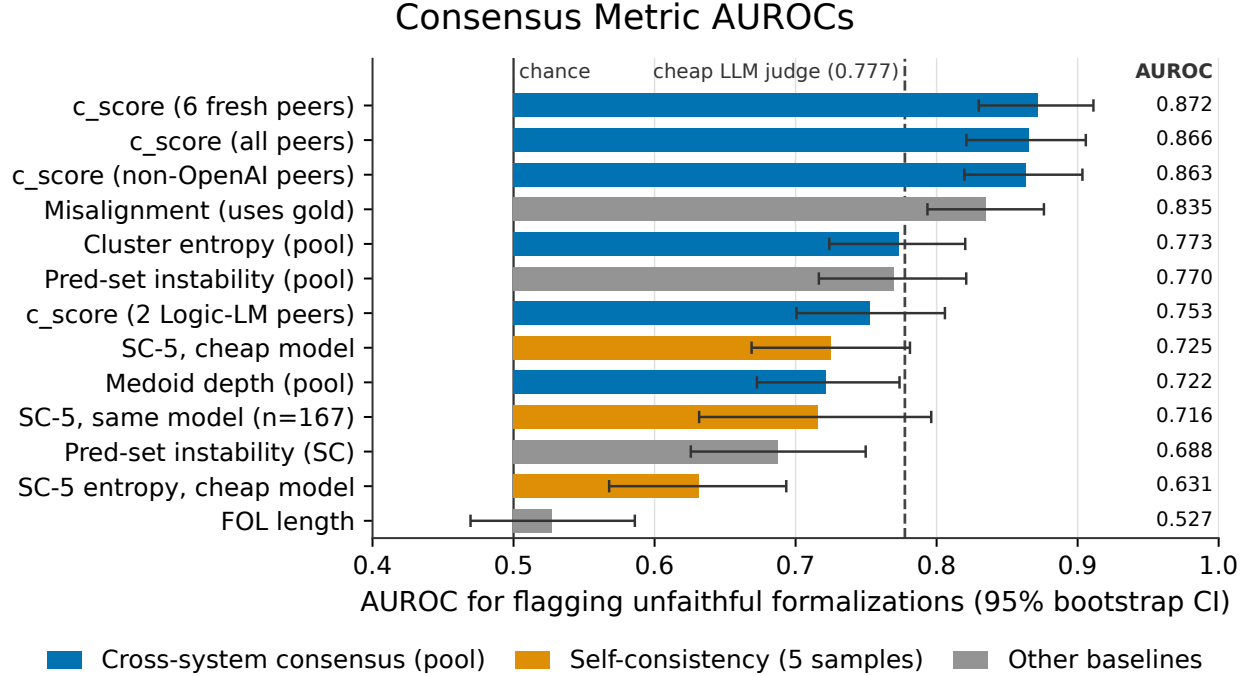


Figure 2: AUROC comparison of consensus-based metrics and baselines on the iteration-1 screen (track L, $n = 546$). Cross-family consensus (c_score) achieves the highest AUROC at 0.866.

As shown in Figure 2, the c_score achieves the highest AUROC in the iteration at 0.866, beating SC-5 (the self-consistency baseline) by $\Delta\text{AUROC} = +0.140$ [0.085, 0.197]. Using only the six fresh peer translations gives 0.872; restricting to the two other Logic-LM systems drops AUROC to 0.753, showing that diversity across model families is essential.

RENAME rewrite invariance: failure. Under predicate renaming, the c_score false-alarm rate is 96% (100% for constant-only renaming). This is fundamental: two formulas with different predicate names are not z3-equivalent.

Blind-spot analysis. 38% of ERROR items are z3-equivalent to the medoid. This places a ceiling on recall of approximately 62%. The blind-spot rate is 5% for polarity errors and 57% for structural errors.

Cost. The peer pool costs \$0.0015 per sentence (6 peers via API). Total experiment spend: \$1.52.

2.4 Experiment D: Disguised Judge and Baselines (gen_art_experiment_4)

This experiment establishes the bar: every baseline metric and every judge variant on the same shared screen, scored with nonce-disguised sentences to control for memorisation. Total API spend was \$2.02.

Figure 3 compares all baseline metrics. **Pilot structural metrics: negative result.** The user’s pilot structural metrics are null on this screen: joint conflict 0.513, arity inconsistency 0.500, shape inconsistency 0.524, dangling predicates 0.533, undeclared predicates 0.525.

Round-trip verification. Round-trip NLI achieves AUROC 0.710 [0.648, 0.769], 0.067 below the cheap judge ($p = 0.008$ by DeLong test). Round-trip reformalisation achieves only 0.513 (76% of items tie).

Table 20: Full baseline table (track L, ERROR vs CORRECT, $n = 524$). Iteration 1, Experiment D.

Metric	AUROC	95% CI
parse_fail	0.500	[0.500, 0.500]
pilot_joint_conflict	0.513	[0.488, 0.539]
pilot_arity_incons	0.500	[0.500, 0.500]
pilot_shape_incons	0.524	[0.470, 0.578]
pilot_dangling	0.533	[0.474, 0.592]
pilot_undeclared	0.525	[0.510, 0.542]
pilot_rerun_jacc	0.720	[0.659, 0.780]
rt_nli_min	0.710	[0.648, 0.769]
rt_nli_fwd	0.652	[0.588, 0.713]
rt_nli_bwd	0.681	[0.621, 0.742]
rt_nli_contra	0.628	[0.566, 0.686]
rt_nli_min_alt	0.689	[0.627, 0.749]
rt_embed_cos	0.633	[0.574, 0.696]
rt_reformalise_eq	0.513	[0.479, 0.547]
judge_cheap_orig	0.749	[0.695, 0.801]
judge_cheap_disg	0.777	[0.720, 0.827]
judge_cheap2_orig (nano)	0.750	[0.691, 0.810]
judge_cheap2_disg	0.714	[0.648, 0.777]
judge_strong_orig (3.1-pro)	0.802	[0.715, 0.877]
judge_strong_disg	0.730	[0.655, 0.803]
judge_local_qwen8b_disg	0.757	[0.702, 0.811]
judge_local_llama8b_disg	0.671	[0.604, 0.737]
judge_local_qwen14b_disg	0.760	[0.681, 0.834]
rt_nli_min_localverb	0.740	[0.680, 0.796]

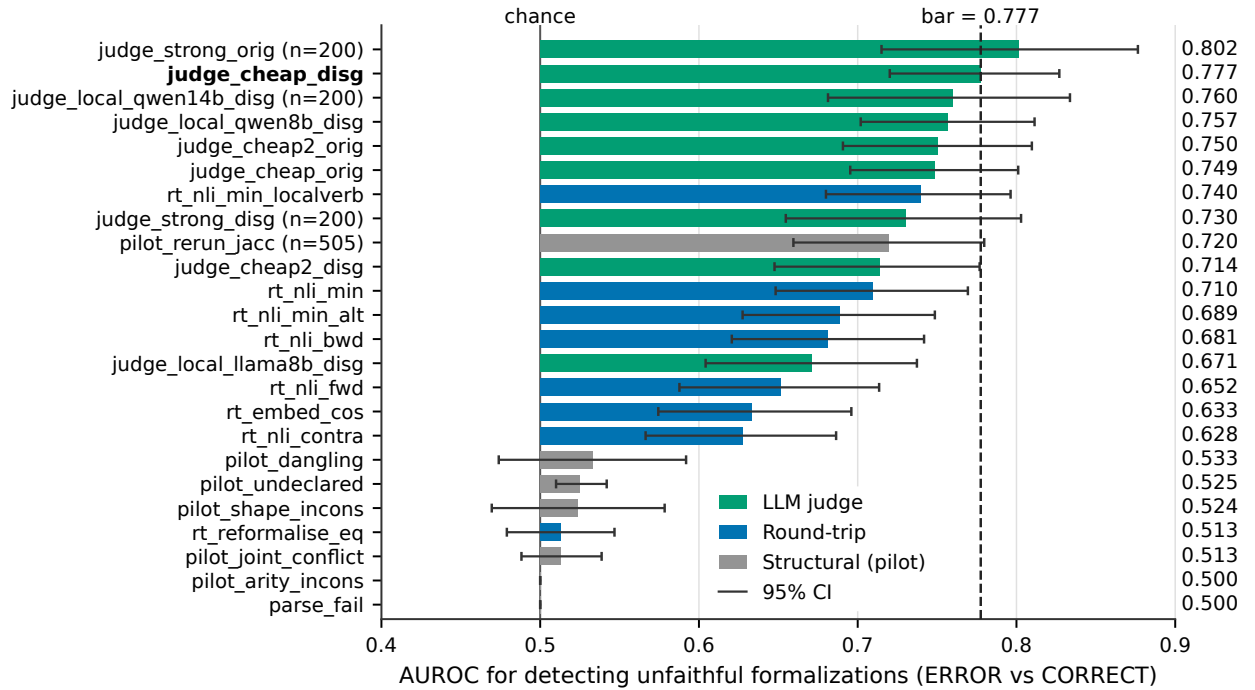


Figure 3: AUROC of all baseline metrics and judge variants on the iteration-1 screen (track L, $n = 524$). The disguised flash-lite judge sets the bar at 0.777.

Frontier judge. The strong judge (Gemini 3.1 Pro Preview) achieves AUROC 0.802 [0.715, 0.877] on the strong subsample ($n = 200$). Under disguise it drops to 0.730. Under panel labels the frontier-judge advantage strengthens to +0.166 [+0.086, +0.244] ($p < 0.001$, $n = 90$).

2.4.1 Cross-fitted baseline combinations (track L)

Table 21: Cross-fitted baseline stacks (track L). Iteration 1, Experiment D.

Combination	AUROC	95% CI
S1 (structural)	0.698	[0.634, 0.761]
S2 (round-trip)	0.687	[0.625, 0.749]
S3 (judges disguised)	0.771	[0.705, 0.831]
S4 (all cheap features)	0.817	[0.762, 0.867]
S6 (S4 + local judges)	0.810	[0.753, 0.861]

2.4.2 Contamination analysis

No evidence of contamination was found. All difference-in-differences CIs include zero:

Table 22: Contamination DiD (H–L) tests. Iteration 1, Experiment D.

Judge	DiD (H–L)	95% CI
flash-lite	−0.042	[−0.170, 0.094]
gpt-4.1-nano	−0.077	[−0.188, 0.023]
gemini-3.1-pro	+0.002	[−0.115, 0.104]
Qwen3-8B	−0.045	[−0.161, 0.079]
Llama-3.1-8B	+0.023	[−0.112, 0.157]
Qwen3-14B	−0.073	[−0.195, 0.056]

2.4.3 Invariance of judges under meaning-preserving rewrites

Judges are highly sensitive to meaning-preserving rewrites. The FA on CONTRAPOSITIVE rewrites is 0.67 for the cheap judge (original) and 0.95 for the cheap2 judge. FOL-Triage’s fused flip rate is 0.013 on RENAME, compared to 0.27–0.49 for judges.

Table 23: Judge invariance: flip rates (track L, 270 CORRECT items). Iteration 1, Experiment D.

Metric	RENAME	REORDER	DEMORGAN	CONTRA	ALL
judge_cheap_orig	0.27	0.12	0.23	0.64	0.29
judge_cheap_disg	0.44	0.12	0.46	0.41	0.39
judge_cheap2_orig	0.49	0.12	0.46	0.95	0.49
rt_nli_min	0.37	0.10	0.31	0.33	0.31
rt_embed_cos	0.57	0.07	0.44	0.38	0.44
FOL-Triage fused	0.013	0.0	0.061	0.0	0.013

2.4.4 Per-error-type sensitivity of judges (track H)

Table 24: Judge per-error-type sensitivity (track H, curator view, $n = 46$ errors). Iteration 1, Experiment D.

Group	n	cheap_orig	cheap_disg	cheap2_orig	rt_nli
1-op errors	13	0.308	0.538	0.462	0.231
2-op errors	11	0.545	0.818	0.727	0.273
ADD	12	0.500	0.750	0.667	0.250
COMPOUND	22	0.545	0.864	0.773	0.273
DROP	11	0.455	0.636	0.455	0.182
CONN	3	0.667	0.667	1.000	0.333
NEG	2	0.500	0.500	1.000	0.500
SWAP	2	0.000	1.000	0.500	0.000

[Correction (C2): On development set E at matched FA 0.10, the original judge has higher recall than the disguised judge (R_SOLVER_CONS diff +0.106 [0.033, 0.178]; R_ADJ_AB diff +0.198 [0.130, 0.260]). Disguise does not uniformly improve the judge.]

2.4.5 Complexity analysis (track L)

Table 25: Judge complexity strata (track L). Iteration 1, Experiment D.

Stratum	n (err/cor)	judge disg AUROC	rt_nli AUROC
words < 12	485 (210/275)	0.784	0.713
words 12–19	36 (18/18)	0.667	0.670
quant. 0–1	512 (226/286)	0.784	0.712
depth ≤ 3	466 (199/267)	0.781	0.702
depth 4–5	54 (30/24)	0.762	0.828
0 conditions	394 (182/212)	0.783	0.745
1–2 conditions	128 (48/80)	0.765	0.653

2.4.6 H_primary label set ($n = 177$, 24 errors)

2.4.7 H_curator_view label set ($n = 199$, 46 errors)

2.4.8 Gold-recall probe

The primary model (Gemini 2.5 Flash Lite) translated each track-H sentence from scratch. Of the 113 items where original gold $\neq$ corrected gold: probe $\equiv$ original public gold on 0.044; $\equiv$ corrected gold on 0.159. Predicate-name Jaccard(probe, original gold) – Jaccard(probe, Logic-LM gpt-4): mean -0.068 $[-0.171, +0.034]$ ($n = 63$).

2.5 Iteration 1 Summary

Figure 4 summarizes the head-to-head comparison. **What iteration 1 established:**

Table 26: H_primary AUROC (track H, solver labels, $n = 177$, 24 errors). `gen_art_experiment_4`.

Metric	AUROC	95% CI
judge_strong_orig	0.878	[0.788, 0.954]
judge_strong_disg	0.798	[0.687, 0.900]
judge_local_qwen8b_orig	0.774	[0.655, 0.874]
judge_cheap_orig	0.753	[0.632, 0.857]
judge_cheap2_orig	0.749	[0.650, 0.840]
rt_nli_min	0.741	[0.633, 0.839]
judge_cheap_disg	0.740	[0.632, 0.839]
judge_local_qwen8b_disg	0.731	[0.615, 0.834]
rt_nli_min_localverb	0.701	[0.583, 0.806]
rt_embed_cos	0.682	[0.569, 0.790]
pilot_dangling	0.672	[0.557, 0.790]
judge_local_llama8b_orig	0.670	[0.535, 0.794]
judge_local_llama8b_disg	0.664	[0.535, 0.779]
judge_cheap2_disg	0.635	[0.517, 0.742]

Table 27: H_curator_view AUROC (track H, curator labels, $n = 199$, 46 errors). `gen_art_experiment_4`.

Metric	AUROC	95% CI
judge_strong_orig	0.915	[0.861, 0.963]
judge_strong_disg	0.846	[0.766, 0.920]
judge_local_qwen8b_orig	0.816	[0.738, 0.888]
judge_cheap_orig	0.806	[0.734, 0.872]
judge_local_qwen14b_orig	0.792	[0.708, 0.865]
judge_cheap2_orig	0.790	[0.717, 0.858]
judge_cheap_disg	0.782	[0.708, 0.849]
rt_nli_min	0.772	[0.696, 0.845]
rt_embed_cos	0.757	[0.678, 0.827]
judge_local_qwen8b_disg	0.755	[0.673, 0.829]
judge_local_llama8b_orig	0.736	[0.645, 0.824]
rt_nli_min_localverb	0.735	[0.652, 0.813]
judge_local_qwen14b_disg	0.727	[0.616, 0.823]
judge_local_llama8b_disg	0.709	[0.621, 0.793]
judge_cheap2_disg	0.688	[0.608, 0.769]
pilot_dangling	0.659	[0.566, 0.756]

Table 28: Iteration 1 summary: all candidates vs the bar. Iteration 1.

Method	AUROC	95% CI	n	\$/item	Flip rate
c_score (consensus)	0.866	[0.821, 0.906]	546	0.0015/sent	0.96 (REN.)
S4 baseline combo	0.817	[0.762, 0.867]	524	varies	varies
Bar: judge disg	0.777	[0.720, 0.827]	524	0.00004	0.39
FOL-Triage fused	0.759	[0.695, 0.825]	477	0.0002	0.013
rt_nli_min	0.710	[0.648, 0.769]	524	0.00001	0.31
L1 lint	0.508	[0.499, 0.519]	477	0	0.0
Pilot structural	0.50–0.53	n/a	524	0	n/a

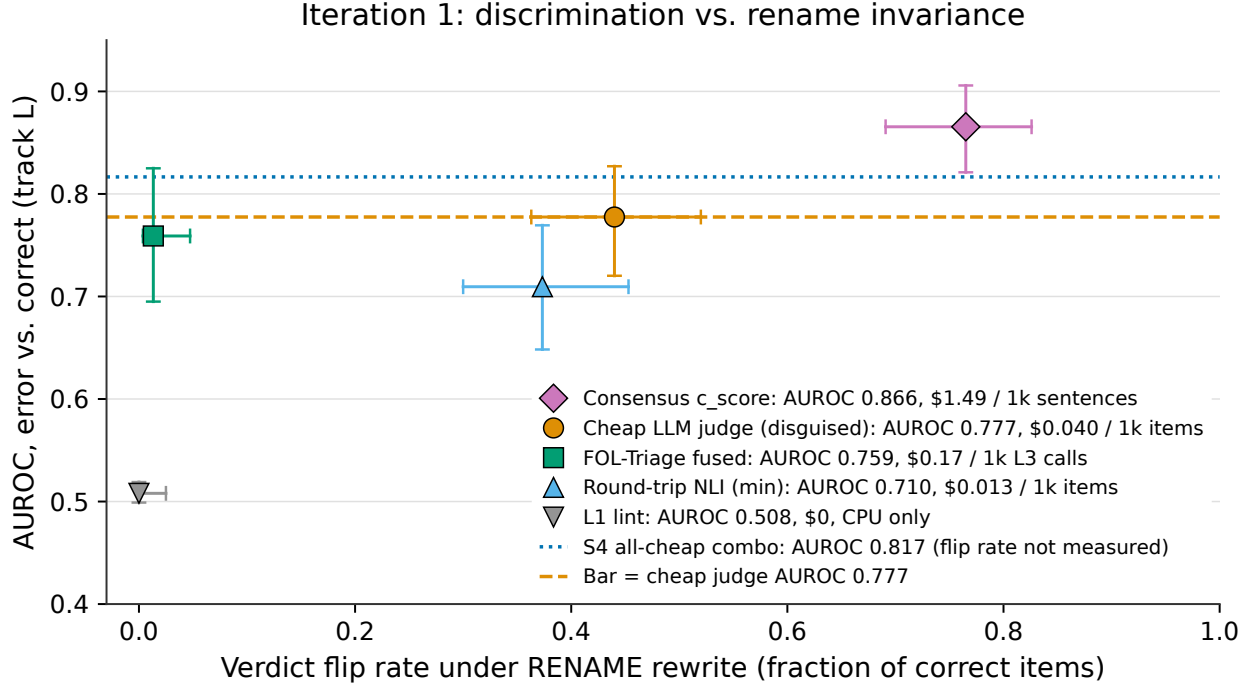


Figure 4: Summary comparison of all metric families at the end of iteration 1. Consensus achieves the highest AUROC (0.866) but has a 96% RENAME false-alarm rate. The disguised cheap judge (0.777) sets the bar.

1. The cross-system consensus metric achieves the highest AUROC at 0.866, beating SC-5 by +0.140. It requires peer translations from multiple model families and is fundamentally non-invariant to predicate renaming (96% FA under RENAME).
2. The disguised cheap judge sets the bar at AUROC 0.777. The S4 combination reaches 0.817 (+0.039, not significant). The frontier judge adds nothing significant over the cheap judge under solver labels; under panel labels the advantage strengthens (+0.166, $p < 0.001$).
3. FOL-Triage achieves AUROC 0.759, below the bar. Its primary value is invariance: the fused flip rate under meaning-preserving rewrites is 0.013, compared to 0.29–0.49 for judges.
4. L1 lint does not transfer from human annotation errors to LLM outputs. The pilot structural metrics are null. Round-trip NLI is 0.067 below the judge.
5. The evaluation labels are noisy: 58.8% of track-L “errors” are pure ADD+DROP (likely vocabulary mismatches).
6. No contamination was detected for any LLM-based metric.

Review score: 3 (blocking). The reviewer identified that the cross-candidate conclusions compared AUROCs from three different label vectors and item sets, that the pre-registered decision rule was never applied (on shared items the paired Δ is not significant), and that multiple statements contradicted artifact evidence.

Hypothesis update: move DEEPEN on the two leads (FOL-Triage text layers; cross-family consensus), merged into one complementary PEER+TEXT fusion.

3 Iteration 2

3.1 Strategy

Iteration 2 has three goals: (i) confirm iteration 1’s ordering on the development dataset E under both the solver labels (R_AB, R_A) and panel-adjudicated labels (R_ADJ); (ii) test whether fusing the consensus signal (PEER) with text-based checks (TEXT = L2-bow + L3) improves over each alone; and (iii) construct a typed perturbation suite (PERTURB) for controlled sensitivity testing.

The pre-registered bar for this iteration is the flash-lite judge on disguised sentences. The shared OpenRouter key hit its daily limit before the development-set experiment completed. The flash-lite bar is therefore UNTESTABLE on E; the fallback local judge (Qwen3-8B, rubric B, greedy; screen AUROC 0.757 vs 0.777 for flash-lite) is the labelled secondary bar.

3.2 Experiment 5: PEER+TEXT on Development Dataset E (gen_art_experiment_5)

PEER+TEXT is a logistic fusion of the consensus signal (c_score with shared-vocabulary alignment, denoted c_score_align) and the text-based FOL-Triage layers (L2-bow, L3 questionnaire). The fusion weights were fitted on the iteration-1 screen and frozen before any E score was computed.

Name-free alignment: failed the pre-registered screen gate. Both name-free variants (NF-pure and NF-anchored) failed the screen gate. A post-hoc analysis with fractional tie-breaking showed the NF-anchored fusion achieves AUROC 0.759 on E (R_AB pooled), 0.031 below the pre-registered PEER+TEXT fusion (0.790), with the CI excluding zero $[-0.042, -0.020]$.

Table 29: PEER+TEXT headline AUROCs on E (R_AB, ERROR vs CORRECT). Iteration 2, Experiment 5.

Subset	<i>n</i> (err/cor)	PT	c_align	NF	L2	L3	Judge
R_AB pooled	2686 (1822/864)	0.790	0.782	0.743	0.734	0.579	0.712
R_AB L25	873 (697/176)	0.755	0.712	0.651	0.722	0.560	0.664
R_AB L20	821 (592/229)	0.747	0.722	0.643	0.715	0.547	0.683
R_AB EXC	606 (384/222)	0.777	0.814	0.782	0.669	0.620	0.663
R_AB CTRL	386 (149/237)	0.708	0.742	0.741	0.613	0.490	0.739
R_AB long	2300 (1673/627)	0.768	0.759	0.704	0.717	0.580	0.653
R_A L20+EXC	449 (261/188)	0.818	0.860	0.802	0.678	0.587	0.554

Figure 5 shows AUROCs across subsets. **PEER+TEXT vs the local judge.** The paired Δ AUROC is +0.078 [0.043, 0.112] on R_AB pooled (DeLong $p < 0.0001$). On the long pool (L25+L20+EXC), the gap widens to +0.115 [0.075, 0.157]. On R_A (tier-A labels, L20+EXC), the gap is +0.264 [0.157, 0.359].

Does fusion beat c_score_align alone? The stratified AUROC difference (PEER+TEXT minus c_score_align) is +0.011 $[-0.012, 0.038]$, not significant. The TEXT layers add no significant discriminative power over c_score_align alone once the stratum-composition confound is removed.

3.2.1 Mechanism predictions

Four pre-registered mechanism predictions were tested on E:

- **P1** (complementary recall): INCONCLUSIVE.
- **P2** (uncorrelated error rankings): REFUTED. Spearman(PEER, TEXT) = 0.294.

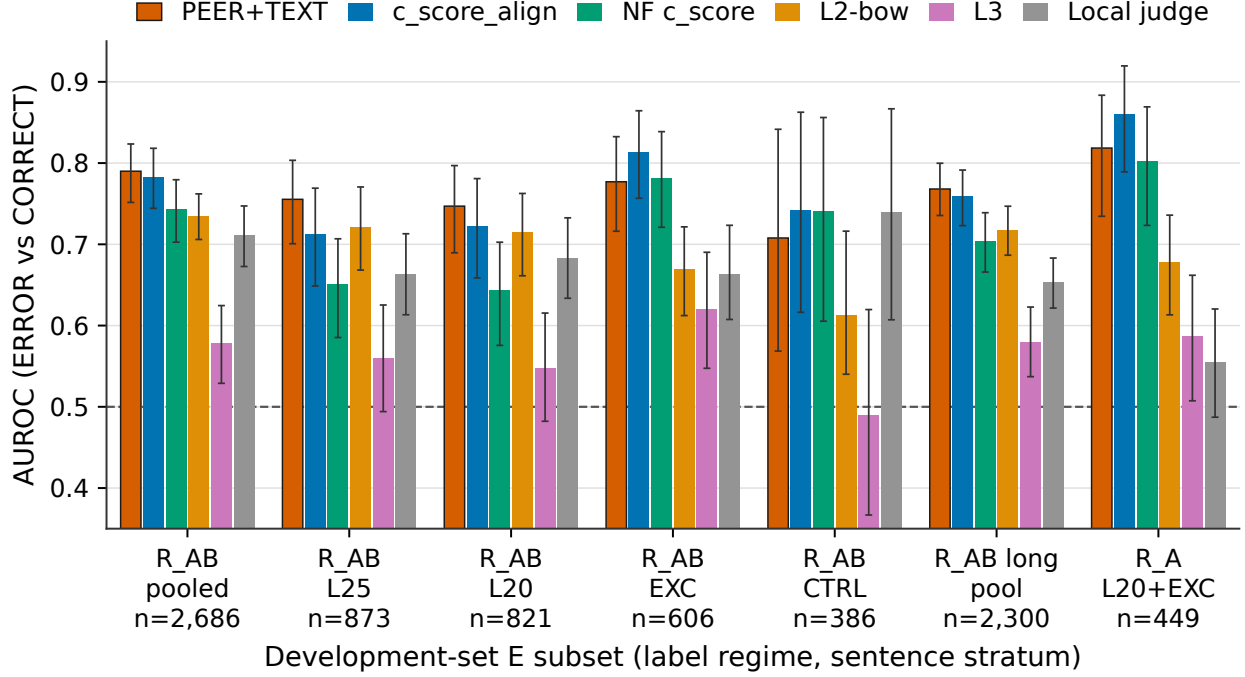


Figure 5: AUROC on development dataset E under R_AB labels. PEER+TEXT leads across all subsets, with the largest advantage on long exception sentences.

- **P3** (complexity interaction): INCONCLUSIVE.
- **P4** (rewrite invariance): REFUTED. Fused RENAME FA is 0.385 (vs FOL-Triage fused FA 0.22).

3.2.2 Rewrite false alarms (screen, CORRECT bases)

Table 30: Rewrite false-alarm rates on screen CORRECT bases. Iteration 2, Experiment 5.

Rewrite	n	Fused	c.align	NF g	L2	L3	Judge
RENAME	104	0.385	0.859	0.074	0.216	0.168	0.683
REORDER	27	0.222	0.178	0.037	0.044	0.444	0.000
DEMORGAN	26	0.154	0.185	0.038	0.046	0.231	0.231
CONTRA.	28	0.107	0.063	0.237	0.075	0.188	0.214

3.3 Evaluation 1: Re-evaluation Under Alternative Label Regimes (gen_art_evaluation_1)

All 57 reviewer-audited quantities from the iteration-1 report reproduce exactly under independent recomputation (57/57 MATCH).

Under R_ADJ_AB, PEER+TEXT (0.877) leads, followed by FOL-Triage fused (0.868). The c_score drops from 0.849 (solver) to 0.779 (R_ADJ_AB), because adjudicated labels reclassify many vocabulary-mismatch items as CORRECT.

Table 31: Regime AUROC matrix (common set, selected metrics). Iteration 2, Evaluation 1.

Metric	R_SOLVER	R_ADJ_AB	R_ADJ_A	R_ADJ_ALL
PT (PEER+TEXT)	0.871	0.877	0.932	0.906
fused_H (FOL-Triage)	0.770	0.868	0.958	0.866
c_score	0.849	0.779	0.786	0.831
L2-bow	0.732	0.815	0.917	0.810
L3	0.761	0.718	0.813	0.751
judge_cheap_disg	0.782	0.772	0.751	0.816
judge_cheap_orig	0.754	0.797	0.858	0.785
sc5_cheap	0.713	0.703	0.676	0.737
rt_nli_min	0.709	0.749	0.731	0.760

3.3.1 Label regime shifts

Table 32: Label-only shift on 161 shared items (36 label changes). Iteration 2, Evaluation 1.

Metric	Shift	95% CI	Direction
fused_H	+0.080	[0.003, 0.171]	UP
L2-bow	+0.102	[0.018, 0.192]	UP
c_score	-0.106	[-0.255, 0.013]	INSIDE CI
judge_cheap_disg	+0.018	[-0.094, 0.134]	INSIDE CI

3.4 Dataset 3: R_COMP and PERTURB (gen_art_dataset_3)

R_COMP: incomplete (OpenRouter key exhausted). The R_COMP component generates long, complex sentences (≥ 25 words, ≥ 3 conditions) by composing atoms from a verified lexicon of 652 entries. Nine templates (T1-T9) produce sentences with weak and strong exception readings. Candidate generation did not run due to the OpenRouter key hitting its daily limit.

PERTURB: complete. The typed perturbation suite applies 12 operator types to 300 base formulas to produce controlled mutants where the error type and direction are known by construction.

Table 33: PERTURB suite statistics. Iteration 2, Dataset 3.

Statistic	Value
Base formulas	300
Total mutants (DOWN + UP)	4,234
Matched controls	868
Total suite rows	5,102
Complete DOWN/UP pairs	1,354
z3 verification failures	0
Operator types	12
Cost	\$0.41

3.5 Experiment 6: Baselines on Development Set E (gen_art_experiment_6)

Experiment 6 scored all local and baseline metrics on dataset E under R_AB labels ($n = 2686$, 1822 ERROR / 864 CORRECT, 292 sentences).

Table 34: Pooled AUROCs on E (R_AB, selected metrics). Iteration 2, Experiment 6.

Metric	AUROC [95% CI]
S4_local (all tiers)	0.754 [0.715, 0.789]
S4_local (R_AB)	0.748 [0.709, 0.782]
judge_local_qwen8b_disg	0.710 [0.674, 0.746]
S4_local_noLLMjudge	0.703 [0.660, 0.742]
sc5_local_eq_frac	0.664 [0.617, 0.706]
pilot_rerun_jacc	0.660 [0.621, 0.695]
judge_local_qwen8b_orig	0.649 [0.604, 0.693]
judge_local_llama8b_disg	0.636 [0.590, 0.681]
rt_nli_min_local	0.626 [0.580, 0.673]
b2_score	0.574 [0.529, 0.619]
pilot_shape_incons	0.504 [0.499, 0.510]
parse_fail	0.500 [0.500, 0.500]

3.6 Dead Ends: B2 and R_ADJ

Candidate B2 (decomposed z3 distinguishing-worlds reader): DROPPED. Gate balanced accuracy on track-H corrected labels was 0.476 (target ≥ 0.85). The b2_score AUROC on E (R_AB pooled) was 0.574, below every other non-null metric.

R_ADJ adjudicator gate: DROPPED.

Table 35: Adjudicator gate results. Iteration 2.

Adjudicator	Gate bal. accuracy	Target
Sonnet-5	0.662	≥ 0.80
Grok-4.20	0.527	≥ 0.80

Inter-adjudicator kappa was 0.497. Sonnet-5 test-retest kappa was 0.834 with a flip rate of 0.07.

3.7 Iteration 2 Summary

PEER+TEXT achieves AUROC 0.790 [0.75, 0.82] on dataset E (R_AB pooled, $n = 2686$), beating the local judge by +0.078 [0.043, 0.112]. Fusion does not significantly beat c_score_align alone (stratified $\Delta = +0.011$ [-0.012, 0.038]). The name-free consensus variant failed the pre-registered gate. Under adjudicated labels the metric ranking changes: FOL-Triage fused rises from 0.770 to 0.868; c_score drops from 0.849 to 0.779. All 57 reviewer-audited numbers reproduce exactly. The PERTURB suite is complete (4,234 typed mutants + 868 controls). R_COMP candidate generation is blocked on the OpenRouter key.

Review score: 3 (blocking). Two executed artifacts were missing from the report (Experiment 6, Dataset 2 R_ADJ). Several statements contradicted artifacts. None of the nine iteration-1 corrections had been applied.

Hypothesis update: move DEEPEN. Title: “Peer agreement vs judges on long sentences.”
Iteration-2 total API spend: \$4.229.

4 Iteration 3

4.1 Strategy

Iteration 3 resolves the two gaps that iteration 2 left open: the flash-lite API bar and the R_COMP candidate generation. It runs five pre-registered tests: T1 (head-on API bar on E), T2 (R_COMP), T3 (PERTURB per-operator sensitivity), T4 (mechanism verification), and T5 (PERTURB rename tradeoff).

4.2 Test T1: Head-on API Bar on Development Set E

The shared OpenRouter key was available for this iteration. All API metrics were scored on E. Population: $n = 2686$ (1822 ERROR / 864 CORRECT, 292 sentences).

Overall T1: CONFIRMED. `c_score_align` (strat AUROC 0.741) exceeds the flash-lite disguised judge (0.642) by $+0.099$ [$+0.049$, $+0.146$].

Table 36: T1 verdict table. Iteration 3.

Criterion	Status	Key numbers
(a) <code>c_align</code> beats flash-lite	CONFIRMED	strat $\Delta +0.099$ [$+0.049$, $+0.146$]
(b) adds signal beyond S4_full	CONFIRMED	strat $\Delta +0.039$ [$+0.021$, $+0.057$]
(d) frontier ratio ≥ 0.95	CONFIRMED	ratio 0.957; $30.2\times$ cheaper
VEX confound control	CONFIRMED	strat $\Delta +0.306$ [$+0.232$, $+0.376$]
M3 length slope	DISCONFIRMED	CI includes zero

Table 37: Item-level AUROCs on R_AB (E_POOL PRIMARY). Iteration 3, Experiment 6.

Metric	Pooled [CI]	Strat [CI]	AUPRC	\$/item
<code>p_peer_text</code>	0.790 [0.752, 0.825]	0.753 [0.720, 0.784]	0.879	1.21e-04
<code>c_score_align</code>	0.782 [0.745, 0.816]	0.741 [0.705, 0.775]	0.861	1.21e-04
<code>S4_full_oof</code>	0.777 [0.739, 0.809]	0.735 [0.699, 0.766]	0.874	—
<code>S4_local_oof</code>	0.748 [0.709, 0.782]	0.693 [0.656, 0.729]	0.856	—
<code>nf_c_score</code>	0.743 [0.702, 0.780]	0.686 [0.647, 0.720]	0.829	—
<code>L2-bow</code>	0.734 [0.705, 0.762]	0.697 [0.667, 0.726]	0.830	—
<code>judge_qwen8b_disg</code>	0.710 [0.674, 0.746]	0.674 [0.646, 0.703]	0.815	—
<code>judge_cheap2_orig</code>	0.700 [0.659, 0.739]	0.653 [0.616, 0.688]	0.823	2.01e-05
<code>judge_cheap_disg</code>	0.696 [0.655, 0.733]	0.642 [0.607, 0.676]	0.786	4.88e-05
<code>sc5_eq_frac</code>	0.656 [0.609, 0.699]	0.598 [0.559, 0.637]	0.758	2.95e-05
<code>rt_nli_min</code>	0.639 [0.593, 0.687]	0.606 [0.569, 0.645]	0.776	2.75e-05
<code>judge_cheap_orig</code>	0.635 [0.595, 0.671]	0.623 [0.593, 0.651]	0.772	4.88e-05
<code>L3</code>	0.579 [0.532, 0.625]	0.562 [0.524, 0.601]	0.720	—
<code>rt_embed_cos</code>	0.574 [0.527, 0.623]	0.566 [0.526, 0.607]	0.731	—
<code>parse_fail</code>	0.500 [0.500, 0.500]	0.500 [0.500, 0.500]	0.678	—

Table 38: Paired Δ vs flash-lite disguised (stratified). Iteration 3.

Cell	n (E/C)	Challenger	Δ [CI]
R_AB pooled	1822/864	c_score_align	+0.099 [+0.049, +0.146]
R_AB pooled	1822/864	p_peer_text	+0.110 [+0.065, +0.154]
R_AB long	1673/627	c_score_align	+0.116 [+0.070, +0.163]
R_AB EXC	384/222	c_score_align	+0.197 [+0.124, +0.269]
R_A L20+EXC	261/188	c_score_align	+0.299 [+0.212, +0.390]
R_AB CTRL	149/237	c_score_align	-0.069 [-0.232, +0.098]

4.2.1 Paired deltas: challenger minus bar (stratified)

On CTRL sentences (short, simple), the consensus advantage vanishes. The advantage is largest on exception sentences (EXC: +0.197) and on R_A tier-A labels (+0.299).

4.2.2 Cross-fitted stacks and nesting

Table 39: Cross-fitted stacks and nested contrasts. Iteration 3.

Stack	Pooled OOF [CI]	Strat OOF [CI]
S4_local	0.748 [0.709, 0.782]	0.693 [0.656, 0.729]
S4_full	0.777 [0.739, 0.809]	0.735 [0.699, 0.766]
S4_full + c_align	0.807 [0.772, 0.838]	0.774 [0.743, 0.802]
S4_full + PT	0.808 [0.775, 0.839]	0.776 [0.745, 0.805]
PT_refit (4 features)	0.808 [0.769, 0.842]	0.778 [0.742, 0.813]

Adding c_score_align to S4_full yields +0.039 strat AUROC (permutation null $p_{95} = 0.009$). The 4-feature PT_refit model matches the full S4 stack without any judge features.

4.2.3 AUROC by complexity bin

Table 40: AUROC by word-count bin. Iteration 3.

Bin	n (E/C)	c_align	PT	judge disg	S4_full	Δ c-judge [CI]
<12	154/283	0.725	0.746	0.766	0.772	-0.040 [-0.178, +0.103]
12-19	255/134	0.701	0.682	0.607	0.667	+0.094 [+0.013, +0.188]
20-24	686/262	0.727	0.740	0.625	0.715	+0.103 [+0.026, +0.171]
25-34	614/161	0.705	0.753	0.618	0.727	+0.087 [-0.006, +0.178]
≥ 35	113/24	0.759	0.756	0.706	0.789	+0.054 [-0.115, +0.173]

Table 41: AUROC by exception type. Iteration 3.

Bin	n (E/C)	c_align	judge disg	Δ [CI]
None	1393/617	0.776	0.731	+0.046 [+0.007, +0.089]
except	62/71	0.964	0.600	+0.365 [+0.236, +0.472]
unless	179/98	0.801	0.713	+0.088 [+0.001, +0.164]

The consensus advantage is concentrated in sentences with 12+ words and is largest on exception-clause sentences (AUROC 0.964 vs 0.600 for “except” sentences).

4.2.4 Recall per error group at matched FA 0.10

Table 42: Recall per error group at matched FA 0.10 (descriptive). Iteration 3.

Group	n	c_align	PT	judge disg	strong	S4_full
COMPOUND (B)	1148	0.55	0.63	0.25	0.67	0.56
polarity	434	0.53	0.56	0.26	0.74	0.53
structural	1282	0.49	0.53	0.24	0.60	0.50
COVERAGE	246	0.27	0.26	0.13	0.27	0.20
MEANING_REN	221	0.07	0.10	0.17	0.09	0.17
PEER_END	398	0.00	0.06	0.16	0.07	0.17
PEER_END_NF	340	0.05	0.11	0.13	0.15	0.21

4.2.5 k -saturation curve

Table 43: k -saturation curve on E. Iteration 3, Evaluation 2.

k	AUROC	AUROC (words T3)	\$/sentence
1	0.685	0.581	\$0.000315
3	0.757	0.651	\$0.000946
5	0.776	0.698	\$0.001576
7	0.784	0.726	\$0.002206

At $k = 3$, the AUROC is 0.757, already 96.6% of the full-pool value (0.784 at $k = 7$). Long sentences need more peers (words-tercile $k_{95} = 2/3/5$).

4.3 Test T2: R_COMP (Long Composed Sentences; gen_art_experiment_7)

R_COMP generates long, complex sentences (≥ 25 words, 4–5 conditions) from a verified lexicon. 221 sentences produced 2,024 candidate translations. Population: 429 ERROR / 1,595 CORRECT.

4.3.1 SIG condition (shared vocabulary)

The consensus metric’s within-template AUROC is 0.954, massively outperforming the flash-lite disguised judge at 0.587. The Δ of +0.367 is the largest effect observed in any test. Error endorsement (e): 0.0—every error is distinguishable from the consensus.

4.3.2 FREE condition: NOT_TESTABLE

The FREE condition produced only 34 CORRECT tier-A items against 420 ERROR. The CI includes zero and the sample is too thin for reliable inference. FREE is NOT_TESTABLE and deferred to iteration 4.

Table 44: SIG per-metric AUROCs (within-template). Iteration 3, Experiment 7.

Metric	n	Within-template [CI]	Pooled [CI]	AUPRC
g_align	2024	0.977 [0.970, 0.983]	0.978 [0.972, 0.983]	0.878
g_nf	2024	0.976 [0.968, 0.983]	0.976 [0.970, 0.982]	0.877
c_score_sig	2024	0.952 [0.937, 0.965]	0.964 [0.957, 0.971]	0.803
c_score_align	2024	0.952 [0.937, 0.965]	0.964 [0.957, 0.971]	0.803
judge.frontier_orig	60	0.932 [0.782, 1.000]	0.932 [0.855, 0.990]	0.881
judge.frontier_disg	60	0.729 [0.573, 0.867]	0.735 [0.593, 0.851]	0.655
judge.cheap_orig	1932	0.673 [0.637, 0.708]	0.710 [0.678, 0.741]	0.345
judge.cheap_disg	1904	0.587 [0.551, 0.622]	0.578 [0.537, 0.617]	0.238
judge.local_disg	2024	0.498 [0.459, 0.534]	0.541 [0.505, 0.575]	0.233

Table 45: Rename invariance on R_COMP. Iteration 3, Experiment 7.

Metric	Unrenamed FA	NONCE FA	SYN FA
c_score_sig	0.260	1.000	1.000
c_score_align	0.260	1.000	0.635
c_score_nf	0.263	0.261	0.252
c_score_hyb	0.260	0.261	0.252

4.3.3 Rename invariance on R_COMP

4.3.4 SIG secondary deltas

Table 46: SIG secondary deltas (within-template). `gen_art_experiment_7`.

Comparison	n	Δ [CI]	p
pooled	1904	0.388 [0.350, 0.427]	0.000
within_sentence	1904	0.369 [0.324, 0.412]	0.000
vs judge_cheap_orig	1932	0.278 [0.239, 0.319]	0.000
vs judge_local_disg	2024	0.454 [0.416, 0.494]	0.000
g_align vs judge	1904	0.390 [0.355, 0.425]	0.000
c_sig vs c_align	2024	0.000 [0.000, 0.000]	1.000

4.3.5 SIG nested adds-signal (cross-fitted logistic)

Table 47: SIG nested adds-signal. `gen_art_experiment_7`.

Model	n	Within-template Δ [CI]	Pooled Δ [CI]
cheap_disg + c_sig	1904	0.377 [0.338, 0.416]	0.401 [0.363, 0.439]
cheap_disg + c_hyb	1904	0.375 [0.337, 0.415]	0.399 [0.360, 0.437]
c_sig + cheap_disg	1904	0.004 [-0.000, 0.008]	0.004 [0.001, 0.006]

4.3.6 SIG frontier subsample (descriptive, $n = 60$)

Cost ratio (frontier \$/call over consensus peer-generation \$/candidate): 6.07.

Table 48: SIG frontier subsample. `gen_art_experiment_7`.

Metric	Pooled AUROC [CI]	Within-template
<code>c_score_sig</code>	0.956 [0.889, 1.000]	0.964
<code>judge_frontier_orig</code>	0.932 [0.855, 0.990]	0.932
<code>judge_frontier_disg</code>	0.735 [0.593, 0.851]	0.729
<code>judge_cheap_disg</code>	0.499 [0.341, 0.653]	0.464
<code>c_score_hyb</code>	0.956 [0.889, 1.000]	0.964

4.3.7 SIG operating points at frozen thresholds

Table 49: SIG operating points. `gen_art_experiment_7`.

Metric	n	Precision	Recall	FA rate	AUPRC	Prevalence
<code>c_score_sig</code>	2024	0.509	1.000	0.260	0.803	0.212
<code>g_align</code>	2024	0.743	0.965	0.090	0.878	0.212
<code>g_nf</code>	2024	0.745	0.965	0.089	0.877	0.212
<code>judge_cheap_disg</code>	1904	0.250	0.795	0.626	0.238	0.207
<code>judge_cheap_orig</code>	1932	0.382	0.686	0.288	0.345	0.206
<code>judge_frontier_orig</code>	60	0.853	0.967	0.167	0.881	0.500

4.3.8 SIG complexity by word tercile and conditions

Table 50: SIG within-template AUROC by word tercile. `gen_art_experiment_7`.

Tercile	n	ERR/COR	<code>c_sig</code>	<code>c_hyb</code>	<code>judge disg</code>	Δ <code>c-bar</code> [CI]
W1	574	125/449	0.959	0.959	0.578	0.382 [0.278, 0.472]
W2	743	149/594	0.947	0.941	0.624	0.331 [0.253, 0.421]
W3	707	155/552	0.953	0.953	0.567	0.388 [0.301, 0.471]

4.3.9 SIG labels per slot and template

Table 51: SIG label distribution per translation slot. `gen_art_experiment_7`.

Slot	CORRECT	ERROR	READ. CHOICE	OFF_SIG	UNPARSE	Error rate
G1	76	118	3	16	8	0.608
G1b	171	23	19	1	7	0.119
G2	195	12	12	1	1	0.058
G3	180	26	15	0	0	0.126
G4	197	4	19	1	0	0.020
G5	140	62	8	9	2	0.307
G6	172	21	21	6	1	0.109
G7	188	24	6	3	0	0.113
G8	192	20	3	5	1	0.094
G9	84	119	10	8	0	0.586

Table 52: SIG label distribution per template. `gen_art_experiment_7`.

Template	CORRECT	ERROR	READ. CHOICE	OFF_SIG	UNPARSE
T1	226	34	0	5	5
T2	199	77	0	3	1
T3	192	15	0	3	0
T4	145	84	0	15	6
T5	188	65	0	6	1
T6	187	40	0	9	4
T7	220	47	0	0	3
T8	71	37	116	6	0
T9	167	30	0	3	0

4.3.10 SIG error types by repair operator

Table 53: SIG error-type AUROC and recall at frozen threshold. `gen_art_experiment_7`.

Op	<i>n</i>	c.sig	c.hyb	judge disg	judge orig
CONN	301	0.941/1.000	0.941/1.000	0.606/0.818	0.653/0.667
NEG	71	0.977/1.000	0.977/1.000	0.581/0.838	0.704/0.677
COMPOUND	54	0.984/1.000	0.968/0.981	0.523/0.809	0.745/0.846
MOVE	43	0.975/1.000	0.975/1.000	0.554/0.811	0.716/0.711
ADD	36	0.961/1.000	0.959/1.000	0.561/0.636	0.708/0.600

4.3.11 SIG system level (10 slots)

Spearman of slot error rate vs mean score: c_score_sig 0.721 (Pearson 0.978); g_align 0.830 (Pearson 0.986); judge_cheap_disg 0.418 (Pearson 0.796).

4.4 Tests T3 + T5: PERTURB Sensitivity and Rename Tradeoff (`gen_art_experiment_8`)

4.4.1 T1: Part A selection (pre-registered rule)

Table 54: T1 Part A variant selection. `gen_art_experiment_8`.

Variant	RENAME FA (i)	(ii) SYN	(iii) NONCE	Eligible	E strat AUROC
HYB	0.050	0.841	0.700	False	0.738
NF-anchored	0.030	0.686	0.485	False	0.686
ALIGN (ref.)	—	—	—	not a candidate	0.741

Neither variant is eligible: no rename-invariant consensus exists here; both are reported as failed.

Table 55: T2: E stratified AUROC per subset. `gen_art_experiment_8`.

Subset	n (err)	c_hyb	c_align	c_nf	g_align	PT
R_AB pooled	2686 (1822)	0.738	0.741	0.686	0.694	0.753
R_AB L25	873 (697)	0.698	0.712	0.651	0.676	0.755
R_AB L20	821 (592)	0.729	0.722	0.643	0.636	0.747
R_AB EXC	606 (384)	0.811	0.814	0.782	0.782	0.777
R_AB CTRL	386 (149)	0.727	0.742	0.741	0.769	0.708
R_AB long	2300 (1673)	0.739	0.741	0.680	0.686	0.757
R_A L20+EXC	449 (261)	0.849	0.855	0.795	0.811	0.819

Table 56: T3: Paired Δ (selected comparisons). `gen_art_experiment_8`.

Subset	HYB–NF	NF–ALIGN	HYB–judge	ALIGN–judge
R_AB pooled	0.052	−0.055	0.060	0.064
R_AB EXC	0.030	−0.032	0.149	0.151
R_AB CTRL	−0.013	−0.001	−0.012	0.003
R_A L20+EXC	0.054	−0.060	0.238	0.243

Table 57: T4: Over-alignment audit on real E pairs. `gen_art_experiment_8`.

Pair group	n pairs	Discordant / both labelled	Disc. rate	Cand. ERROR share
ALIGN_agree	7459	447/3521	0.127	0.429
NEW_hyb_not_align	686	69/178	0.388	0.704
DISAGREE_hyb_false	28815	3243/7994	0.406	0.773

4.4.2 T2: E stratified AUROC per subset

4.4.3 T3: Paired Δ stratified AUROC

4.4.4 T4: Over-alignment audit

4.4.5 T5: Over-alignment probes (recall at frozen threshold)

Table 58: T5: Over-alignment probes. `gen_art_experiment_8`.

Probe	c_align	c_nf	c_hyb	g_hyb
screen MEANING_RENAME ($n = 150$)	0.987	0.198	0.247	0.130
screen ROLE_PERMUTE ($n = 131$)	1.000	0.957	0.962	0.690
screen SWAP ($n = 150$)	0.973	0.940	0.947	0.593
perturb MR_DOWN ($n = 169$)	1.000	0.614	0.811	0.218
perturb MR_UP ($n = 149$)	0.993	0.593	0.765	0.219
perturb SWAP_DOWN ($n = 63$)	1.000	1.000	1.000	0.678
perturb SWAP_UP ($n = 53$)	1.000	1.000	1.000	0.491

4.4.6 T6: Complexity by bin on E (R_AB)

c_nf achieves within-base AUROC 0.499 (chance) on MEANING_RENAME, meaning it is blind to this error class. c_align achieves 0.872. Round-trip NLI is complementary: strongest on ADD

Table 59: T6: AUROC by complexity bin. `gen.art_experiment_8`.

Dimension	Bin	n (err)	c_hyb	c_align	c_nf	HYB–ALIGN [CI]
words	<12	437 (154)	0.718	0.725	0.751	−0.008
words	12–19	389 (255)	0.694	0.701	0.666	−0.007
words	20–24	948 (686)	0.731	0.727	0.657	0.004
words	25–29	395 (288)	0.723	0.723	0.685	0.000
words	30+	517 (439)	0.663	0.683	0.614	−0.019
n_cond	0–1	566 (256)	0.789	0.794	0.786	−0.005
n_cond	2	200 (122)	0.650	0.662	0.689	−0.012
n_cond	3	406 (280)	0.798	0.785	0.787	0.013
n_cond	4+	1514 (1164)	0.718	0.724	0.640	−0.006
exception	exc	676 (429)	0.808	0.811	0.777	−0.004
exception	none	2010 (1393)	0.773	0.776	0.732	−0.003

Table 60: T3: Per-operator within-base AUROC (E_bases, selected metrics). Iteration 3, Experiment 8.

Operator	n	c_align	c_hyb	c_nf	rt_nli	judge disg
NEG	379	0.869	0.884	0.843	0.873	0.723
REV	179	0.860	0.869	0.826	0.578	0.697
QUANT	179	0.860	0.872	0.829	0.351	0.632
RESTR	177	0.859	0.873	0.827	0.608	0.638
CONN	327	0.861	0.870	0.832	0.563	0.663
DROP	191	0.841	0.841	0.791	0.792	0.728
ADD	472	0.859	0.860	0.820	0.895	0.683
SWAP	116	0.882	0.910	0.863	0.454	0.529
BIND	314	0.865	0.858	0.798	0.833	0.672
MEAN_REN	318	0.872	0.685	0.499	0.930	0.600

(0.895) but blind to QUANT (0.351) and SWAP (0.454).

Table 61: T5: Rename tradeoff table (E_bases). Iteration 3, Experiment 8.

Metric	NONCE FA	SYN FA	MR wb-AUC	SWAP rec.	Base FA
c_align	0.765	0.565	0.872	0.758	0.201
c_nf	0.132	0.201	0.499	0.512	0.163
c_hyb	0.216	0.298	0.685	0.832	0.197
g_align	0.413	0.318	0.841	0.426	0.154
g_nf	0.132	0.176	0.477	0.428	0.154
rt_nli	0.686	0.191	0.930	0.153	0.143
judge disg	0.229	0.132	0.600	0.220	0.129
S4_local	0.804	0.383	0.760	0.758	0.286

The rename tradeoff is stark: c_align has RENAME_NONCE FA of 0.765 (76.5% of correctly translated formulas flagged after renaming), while c_nf has 0.132 but sacrifices SWAP recall (0.512 vs 0.758). No consensus variant achieves both low rename FA and high structural-error recall.

4.5 Test T4: Mechanism Verification (gen_art_evaluation_2)

48 numerical claims from the iteration-2 report were independently recomputed: 48/48 VERIFIED_MATCH.

Mechanism predictions: M1 (endorsement slope) INCONCLUSIVE. M2 (divergence rises with words) CONFIRMED but non-specific (shuffled-label placebo passes). M3 (complexity slope) INCONCLUSIVE (two specifications conflict). M4 (k -saturation) CONFIRMED ($k_{95} = 3$; by words tercile $k_{95} = 2/3/5$).

4.6 Prior Art Positioning (gen_art_research_1)

A prior-art search (63 sources, 139 verbatim quotes, \$0 LLM spend) positioned the iteration-3 claim that cross-family solver consensus is a gold-free NL-FOL faithfulness metric. The scoop rule—multi-family translations + solver equivalence + faithfulness-label meta-evaluation on NL-FOL—was not met by any prior work.

Table 62: Claim verdicts from prior-art positioning. gen_art_research_1.

Claim	Verdict
C1 (beats API judge; adds over stack)	NEEDS-QUALIFIER: method not new; claim first NL-FOL cross-family meta-evaluation
C2 (holds with aligner-free labels)	SAFE: no neighbour controls for shared instrument
C3 (rename-invariant hybrid matcher)	NEEDS-QUALIFIER: alignment exists in LogicLLaMA/Vossel; rename FA not reported by others
C4 (e/d decomposition)	NEEDS-QUALIFIER: scatter premise stated by others; new piece is measured endorsement vs conditions curve
C5 (3–5 families suffice)	NEEDS-QUALIFIER: already recommended by LLMs-as-a-Jury; novelty per complexity tercile only

Table 63: Closest neighbours from prior-art search. `gen_art_research_1`.

System	Domain	Key result
ARc	NL→SMT, fixed schema	Same score form as <code>c_score</code> ; downstream QA labels, not faithfulness
NoTB	RTL, 4 families	Precision 63/85.3/87/94.7% at ≥ 1.4 families; no AUROC
GenV	NL→FOL, trained verifier	AUROC 0.961; SC $K = 5$ baseline 0.863; judge wins under panel labels
SCP-NL2TL	NL→temporal logic	SC AUROC rises with difficulty tier; closest meta-eval design

4.7 Iteration 3 Summary

T1 CONFIRMED: `c_score_align` beats the API bar (+0.099 [+0.049, +0.146] strat AUROC). Adding `c_score_align` to `S4_full` yields +0.039 [+0.021, +0.057] (permutation null $p_{95} = 0.009$).

T2 NOT CONFIRMED. SIG achieves within-template AUROC 0.954 ($\Delta = +0.367$ vs judge), but SIG is the regime the user excluded. FREE is NOT TESTABLE.

T3–T5: Per-operator sensitivity profiled. Rename tradeoff quantified. No consensus variant achieves both low rename FA and high structural-error recall.

Dead ends confirmed: B2 DROPPED (gate 0.476 vs target 0.85). R_ADJ DROPPED (Sonnet-5 balanced accuracy 0.662).

Review score: 3 (blocking). Hypothesis update: move DEEPEN. Title: “Peer agreement in the candidate’s own words.” Iteration-3 total API spend: \$4.33.

5 Iteration 4

5.1 Strategy

Iteration 4 tests Candidate-Signature Consensus (CSC): instead of having peers translate independently or from a shared external vocabulary, CSC feeds each peer the candidate formula’s own predicate symbols and asks it to translate using those symbols. In parallel, the iteration constructs the E2 confirmation dataset (550 fresh sentences), generates name-free labels for R_COMP FREE candidates, and runs a vocabulary-analysis evaluation.

5.2 Experiment 9: CSC on Development Set E—Provisional Negative

CSC was tested on 354 rows from E (196 ERROR / 158 CORRECT, 144 sentences). The budget event at \$0.108 stopped generation early.

5.2.1 Anchoring: the core finding

Table 64: CSC anchoring: *e* and *d* components. Iteration 4, Experiment 9.

Component	CSC	FREE_exact (matched)
<i>e</i> (error endorsement)	0.459	0.173
<i>d</i> (correct-row divergence)	0.139	0.323

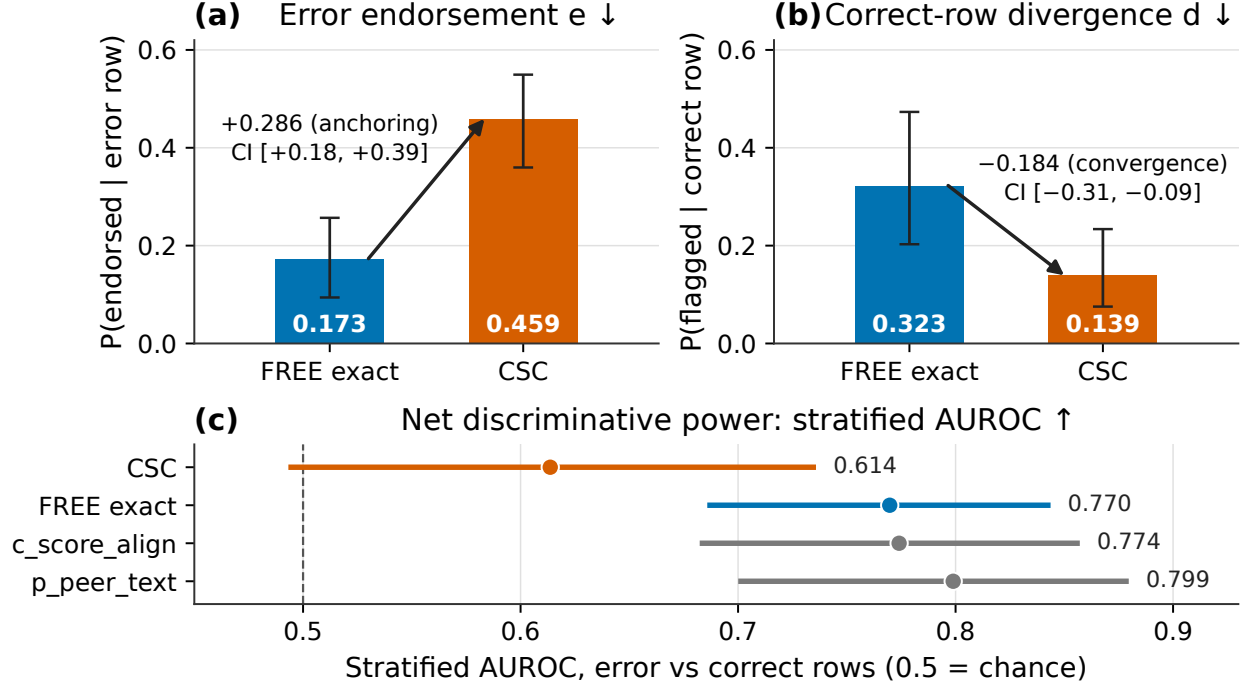


Figure 6: Error endorsement (e) and correct-row divergence (d) under Candidate-Signature Consensus (CSC) vs FREE consensus on development set E. CSC lowers d but raises e dramatically, producing a net negative effect on discriminative power.

CSC lowers d (correct translations converge when given the candidate’s vocabulary) but raises e from 0.173 to 0.459 (nearly half of all errors are endorsed by anchored peers), as shown in Figure 6. The net effect is negative.

5.2.2 Headline AUROCs

Table 65: CSC headline AUROCs (E, PRIMARY, stratified). Iteration 4, Experiment 9.

Metric	Strat AUROC [CI]
p_peer_text	0.799 [0.700, 0.880]
c_score_align	0.774 [0.682, 0.857]
FREE_exact	0.770 [0.686, 0.844]
S4_full	0.736 [0.635, 0.826]
flash-lite judge	0.664 [0.568, 0.750]
CSC	0.614 [0.493, 0.736]

CSC is below every established metric. Gates G1 and G2 fail. The anchoring effect is concentrated in MEANING.RENAME ($e_{\text{CSC}} = 0.821$ vs $e_{\text{FREE}} = 0.036$) and ADD errors ($e_{\text{CSC}} = 0.490$ vs $e_{\text{FREE}} = 0.135$).

Table 66: Anchoring by error class. Iteration 4, Experiment 9.

Error class	e_{CSC}	e_{FREE}
MEANING.RENAME	0.821	0.036
ADD	0.490	0.135
COMPOUND	0.282	0.107

5.3 Experiment 10: CSC on PERTURB—Budget Exhausted (gen_art_experiment_10)

T0 status: the pre-registered CSC peers could not be generated. Every OpenRouter call returned HTTP 403 (run-wide budget of \$7.00 exhausted: \$7.03 at 07:26 UTC). This artifact spent \$0.0000088 (one 1-token probe). Gates G3-P, G4 and MT(E) are UNTESTED. Zero-cost arms were scored instead.

The run-wide OpenRouter budget (\$7.00) was exhausted before experiment 10’s first CSC peer call. Zero-cost arms were scored instead.

SIGPROXY (3-family cued consensus on R_COMP): within-template AUROC 0.930, reaching 98% of the full SIG consensus (0.952).

Table 67: SIGPROXY and controls on R_COMP. Iteration 4, Experiment 10.

Metric	n	Within-template AUROC [CI]
c_proxy (K3)	2024	0.930 [0.916, 0.943]
c_proxy_graded	2024	0.938 [0.926, 0.950]
c_score_sig (full)	2024	0.952 [0.936, 0.966]
g_align	2024	0.977 [0.970, 0.983]
judge_cheap_disg	1904	0.587 [0.550, 0.624]
judge_local_disg	2024	0.498 [0.462, 0.534]

Error typing requires shared vocabulary. Same-vocabulary typed repair achieves 0.784–0.921 accuracy; independent-vocabulary typing drops to 0.085.

Table 68: Same-vocabulary typed repair accuracy. Iteration 4, Experiment 10.

Target	Base source	n	Strict accuracy [CI]
oracle (known vocab)	ALL	4234	0.921 [0.915, 0.928]
proxy (SIGPROXY)	ALL	1196	0.784 [0.711, 0.849]
free3 (indep. vocab)	E only	3884	0.123 [0.081, 0.169]

5.3.1 T1: Signature sharing by operator

5.3.2 T2: Mutant endorsement e per operator

5.4 Dataset 4: E2 Confirmation Set—Partial (gen_art_dataset_4)

E2 is a fresh 550-sentence confirmation set (350 L25, 100 EXC, 100 DT). **Status: PARTIAL (budget-stopped).** Only a 30-sentence pilot (300 candidate rows) completed. All confirmation cells have zero CORRECT rows. Projected cost to complete: \$6.79.

Table 69: T1: Which PERTURB operators share the base signature. `gen_art_experiment_10`.

Operator	Share base sig	Category
BASE, NEG, REV, QUANT, RESTR	1.0	Signature-preserving
CONN, SWAP, MOVE, DEMORGAN	1.0	Signature-preserving
CONTRAPOSITIVE, REORDER_*	1.0	Signature-preserving
ADD_INTERNAL	1.0	Signature-preserving
BIND (E)	0.094	Partly new
DROP (E)	0.037	Partly new
ADD_FOREIGN	0.0	Always new
MEANING_RENAME	0.0	Always new
RENAME_NONCE, RENAME_SYN	0.0	Always new

Table 70: T2: Mutant endorsement (e , share endorsed at $c \leq 0.5$), E bases. `gen_art_experiment_10`.

Op	LOCAL2_own	LOCAL2_base	FREE_align	FREE3_exact	Judge disg
ADD_FOREIGN	0.365	0.269	0.000	0.000	0.078
ADD_INTERNAL	0.294	0.294	0.000	0.000	0.035
MEANING_REN	0.388	0.265	0.000	0.000	0.298
DROP	0.519	0.296	0.010	0.000	0.224
SWAP	0.467	0.467	0.000	0.000	0.300
NEG	0.264	0.264	0.000	0.000	0.070
ALL_MUTANTS	0.341	0.276	0.001	0.000	0.124

Table 71: E2 composition: active sentences per stratum. `gen_art_dataset_4`.

Stratum	Active sent.	With candidates	Mean words	Mean n_cond
L25	350	12	28.8	4.65
EXC	100	8	20.5	3.31
DT	100	10	17.1	2.0

Table 72: E2 pilot label counts (30-sentence, 300 LLM candidate rows). `gen_art_dataset_4`.

Stratum	Final label	Tier	Rows
DT	ERROR	A_unaudited_ref	21
DT	UNPARSEABLE	–	8
DT	UNRESOLVED	none	71
EXC	ERROR	A_unaudited_ref	15
EXC	UNPARSEABLE	–	1
EXC	UNRESOLVED	none	64
L25	UNPARSEABLE	–	11
L25	UNRESOLVED	none	109

5.5 Dataset 5: R_COMP FREE Name-Free Labels

Labels are search-only (gloss step not run). ERROR_CERT (no map is z3-equivalent): 759. MAPPED (at least one map is z3-equivalent but name meaning unverified): 1,506. The CORRECT class is empty, so FREE AUROC is NOT_TESTABLE.

SIG replay validation: 0/1,595 false errors. ERROR_CERT precision: 0.867 [0.758, 0.931] (below the 0.90 flag). Iteration-3 FREE tier-A ERROR labels are 70.7% MAPPED, confirming the aligner-based labeller over-calls errors.

5.6 Evaluation 3: Vocabulary Analysis (gen.art_evaluation_3)

64 claims from the iteration-3 report were checked against source files: 60 match, 1 mismatch (exp-7 peer generation \$/candidate: stated \$0.000996 vs source \$0.000116), 0 NOT_IN_FILES. The mismatch concerns a cost unit (per-candidate vs per-sentence).

Table 73: Cross-family agreement floor (3-pool PRIMARY). Iteration 4, Evaluation 3.

Statistic	Value
END_MAJ d	0.415
Predicted d_{floor}	0.402 [0.336, 0.471]
Bracket	[0.396, 0.415]
R_exact AUROC	0.683
No-anchoring oracle floor	0.385
e ceiling	0.162

SIG-minus-FREE cross-family exact-agreement drop: 0.479 [0.447, 0.513]. The Part-1 model is NOT VALIDATED against FREE data (calibration error -0.372). L25 MDE80 at 350 sentences: 0.112.

5.7 Iteration 4 Summary

CSC is a provisional negative: strat AUROC 0.614, anchoring $e = 0.459$. Gates G1 and G2 fail. Cued consensus confirms k -saturation (SIGPROXY 0.930 at $k = 3$). Vocabulary sharing reduces d by 0.61–0.84 on R_COMP. E2 is constructed but not testable. R_COMP FREE labels are search-only, gloss pending.

Dead ends accumulated:

Table 74: Dead ends across all iterations (through iteration 4).

Dead end	Iter.	Reason
L1 lint on LLM outputs	1	AUROC 0.508
L2 role-aware	1	AUROC 0.557 (below bag-of-words)
Pilot structural metrics	1	AUROC 0.500–0.533 (null)
Round-trip reformalisation	1	AUROC 0.513 (76% ties)
B2 decomposed z3 reader	2	Gate balanced accuracy 0.476
R_ADJ LLM adjudicator	2	Sonnet-5 balanced accuracy 0.662
Name-free consensus (c_nf)	2–3	SWAP recall 0.512
CSC (provisional)	4	Strat AUROC 0.614, anchoring $e = 0.459$

Review score: 3 (blocking). Hypothesis update: move DEEPEN. Title: “Cross-model agreement, tested on fresh data.”

6 Iteration 5

6.1 Strategy

Iteration 5 is the final iteration. It attempts four things: (1) run the E2-A confirmation on 110 untouched sentences; (2) extend E2 to long sentences (E2-B, 400 sentences ≥ 25 words); (3) complete R_COMP FREE confirmation by running the gloss gate; and (4) freeze V0 and screen five consensus variants (V1–V5) plus a Gloss-Gated (GG) name-free checker.

6.2 Experiment 11: E2-A Confirmation (`gen_art_experiment_11`)

E2-A is the untouched portion of the E2 confirmation set: 110 sentences (37 L25, 37 EXC, 36 DT).

Overall verdict: NOT_TESTABLE. The pre-registered flash-lite disguised bar has zero scored rows in R_AB.

6.2.1 DT stratum (testable)

The DT cell (179 rows: 74 ERROR / 105 CORRECT, 26 sentences) is testable. V0 strat AUROC 0.916 [0.858, 0.970]—the highest single-cell consensus AUROC on any confirmation data. The DT gold is trusted by construction (Prover9-validated FOL verbalised by an LLM).

Table 75: E2-A DT stratum AUROCs. Iteration 5, Experiment 11.

Metric	DT strat AUROC [CI]
c_score_align (V0)	0.916 [0.858, 0.970]
p_peer_text	0.919 [0.860, 0.976]
S4_E2	0.953 [0.911, 0.986]
judge_local_qwen8b_disg	0.913 [0.858, 0.958]
l3_z3	0.572 [0.480, 0.675]

6.2.2 Long pool (L25 + EXC, not testable)

The long pool has 94 rows (76 ERROR / 18 CORRECT, 12 sentences): not testable. V0 strat AUROC 0.603 [0.463, 0.735]. L25 alone (42 rows, 7 CORRECT): V0 strat AUROC 0.443 [0.375, 0.500], near chance. Figure 7 maps all confirmation arms and their outcomes.

6.3 Experiment 12: E2-B (Long-Sentence Pool)

Workspace: `gen_art_experiment_12`.

E2-B extends E2 with 400 sentences ≥ 25 words (98 from the E2 surplus pool, 302 continuation; mean 26.82 words, mean 4.66 conditions; word bins 25–29: 374, 30–34: 26; E2 pool reproduction verified, all collision checks zero). **Status: NOT_TESTABLE.** The panel did not complete a single batch before the run-level budget was exhausted. Labels are solver-vs-unaudited-MALLS-gold only (R_A_UNAUDITED). Label counts: 48 CORRECT, 334 ERROR, 3,151 UNRESOLVED, 467

Iteration-5 confirmation arms: status, AUROC and verdict

Arm (labelled set)	Status	Consensus V0 AUROC [95% CI]	Judge (disguised) AUROC [95% CI]	Pre-registered verdict	Arm spend
E2-A 110 / 550 sentences panel-labelled	DT: testable 179 rows 74 ERR / 105 COR	0.916 [0.858, 0.970] n = 179	0.913 [0.858, 0.958] Qwen3-8B · n = 179	NOT_TESTABLE flash-lite judge bar missing	\$2.05
	L25: not testable 42 rows 35 ERR / 7 COR	0.443 [0.375, 0.500] n = 42	0.659 [0.450, 0.967] Qwen3-8B · n = 42		
E2-B 400 sentences no panel labels	48 CORRECT < 50 334 ERR / 48 COR (vs unaudited gold)	0.737 [0.645, 0.827] exploratory, n = 381	0.496 [0.356, 0.620] flash-lite · n = 276	NOT_TESTABLE CORRECT floor not met	\$0.83
R_COMP FREE 2,265 labelled rows	Gloss gate FAILED ×2 759 ERR / 1,506 MAPPED → 0 CORRECT	0.801 [0.770, 0.834] provisional, n = 1,706	0.584 [0.550, 0.618] flash-lite · n = 1,706	NOT_TESTABLE 0 CORRECT rows (fallback B)	\$1.97

Shared \$12.00 OpenRouter budget
exhausted (~11:40 UTC)

Testable cell
 Exploratory / provisional (cannot confirm)
 Not testable

Figure 7: Iteration 5 confirmation attempts and their outcomes. All three pre-registered confirmation arms were budget-stopped before reaching testability. The one testable cell (E2-A DT) shows strong consensus performance (0.916) but is not the target stratum.

UNPARSEABLE. No cell is testable under E’s pre-registered rule (≥ 50 ERROR, ≥ 50 CORRECT, ≥ 25 sentences each): R_A_UNAUDITED has only 48 CORRECT rows.

Table 76: E2-B exploratory AUROCs, all metrics (R_A_UNAUDITED, L25 cell). gen_art_experiment_12. NOT TESTABLE.

Metric	Strat AUROC [CI]	Pooled	AUPRC	Tie rate	n (ERR/COR)
V0 (c_score_align)	0.737 [0.645, 0.827]	0.742	0.949	0.065	381 (333/48)
p_peer_text	0.655 [0.546, 0.767]	0.659	0.930	0.000	382 (334/48)
c_exact	0.678 [0.564, 0.779]	0.675	0.921	0.324	381 (333/48)
V1 (plurality-norm.)	0.733 [0.654, 0.808]	0.740	0.940	0.269	381 (333/48)
V2 (reliability-wt.)	0.750 [0.659, 0.838]	0.755	0.954	0.030	381 (333/48)
V3 (V1+V2)	0.761 [0.681, 0.835]	0.767	0.947	0.215	381 (333/48)
V4 (two-channel log.)	0.764 [0.666, 0.851]	0.766	0.954	0.039	381 (333/48)
V5 (V3 on 3-pool)	0.759 [0.676, 0.834]	0.763	0.944	0.249	378 (330/48)
flash-lite disguised	0.496 [0.356, 0.620]	0.491	0.886	0.259	276 (245/31)
S4_E2B	0.498 [0.402, 0.594]	0.498	0.888	0.001	382 (334/48)
S4_E2B + V0	0.618 [0.487, 0.754]	0.622	0.908	0.000	382 (334/48)
l2_bow	0.603 [0.492, 0.709]	0.605	0.909	0.132	382 (334/48)
l3_z3	0.442 [0.344, 0.552]	0.443	0.867	0.167	382 (334/48)
pilot_joint_conflict	0.500 [0.500, 0.500]	0.500	0.874	1.000	382 (334/48)

Verdicts. B1 (V0 – flash-lite): NOT TESTABLE (E’s rule); exploratory CI > 0. B2 (nested [S4+V0] – S4): NOT TESTABLE; exploratory CI > 0. B3 (V0 – cost-matched): NOT RUN

Table 77: E2-B paired Δ AUROC (R_A_UNAUDITED). `gen_art_experiment_12`.

Contrast	Δ	95% CI	n (COR, both-class sent.)
V0 – flash-lite disg	+0.208	[+0.066, +0.363]	276 (31, 17)
p_peer_text – flash-lite disg	+0.183	[+0.045, +0.334]	276 (31, 17)
[S4_E2B + V0] – S4_E2B (nested)	+0.120	[+0.013, +0.237]	382 (48, 24)
V0 – S4_E2B	+0.238	[+0.111, +0.366]	381 (48, 24)
V1 – V0 (dev-only)	−0.004	[−0.053, +0.051]	381 (48, 24)
V2 – V0 (dev-only)	+0.013	[+0.002, +0.026]	381 (48, 24)
V3 – V0 (dev-only)	+0.024	[−0.025, +0.077]	381 (48, 24)
V4 – V0 (dev-only)	+0.027	[−0.026, +0.081]	381 (48, 24)
V5 – V0 (dev-only)	+0.022	[−0.043, +0.085]	378 (48, 24)
c_exact – V0 (dev-only)	−0.059	[−0.163, +0.033]	381 (48, 24)
V0 – flash-lite vfb (T1 sens.)	+0.206	[+0.062, +0.363]	276 (31, 17)
V0 – flash-lite (judge-ok only)	+0.206	[+0.054, +0.362]	264 (29, 16)

(HTTP 403). B4 (frontier): NOT TESTABLE ($n = 2$). H_IMPROVE: no carried variant (selection winner = “NONE: V0 stands”; V1–V5 reported as development-only rows). H_MECH: (i) UNINFORMATIVE (tautological under gold-equivalence labels); (ii) UNINFORMATIVE; (iii) REFUTED; (iv) PARTIAL.

Label coupling: removing gold-equivalent peers from V0’s pool, V0_decoupled drops to strat AUROC 0.611 [0.490, 0.738] vs V0 0.737; V0_decoupled minus flash-lite disguised +0.133 [−0.024, +0.314].

Rename invariance (E2-B, $c > 0.5$). RENAME_SYN: $n = 31$, FA 0.968, base FA 0.839, paired flip 0.129. RENAME_NONCE: $n = 48$, FA 1.000, base FA 0.833, paired flip 0.167.

Cost. V0 FULL: \$0.000859 per candidate (9 peer generations); MARGINAL \$0 (peers are other systems’ outputs); z3: 0.059s per candidate. Flash-lite disguised: \$5.6e−5 per call.

System-level τ_b . V0: 0.467 [0.055, 0.778]; p_peer_text: 0.422 [−0.022, 0.637]; flash-lite disguised: −0.315 [−0.452, 0.230]; S4_E2B: 0.200 [−0.333, 0.455] (10 systems, descriptive only).

Placebo. Shuffled-peers V0 AUROC: 0.500 (vs same-rows 0.758), confirming the signal is not an artefact of the score form.

Table 78: E2-B coverage per generator system (label-free). `gen_art_experiment_12`.

Slot	System	Parse rate	V0 scorable	Flash-lite ans.	Mean V0
G1	llama-3.1-8b-instruct	0.762	0.762	0.585	0.903
G1b	llama-3.3-70b-instruct	0.902	0.900	0.693	0.839
G2	qwen3-235b-a22b-2507	0.795	0.795	0.613	0.846
G3	mistral-small-3.2-24b	0.920	0.920	0.713	0.847
G4	deepseek-v3.2	0.907	0.905	0.723	0.831
G5	gemma-3-27b-it	0.922	0.917	0.715	0.892
G6	phi-4	0.907	0.905	0.705	0.825
G7	gpt-4.1-mini	0.875	0.875	0.682	0.818
G8	gemini-2.5-flash	0.940	0.935	0.720	0.859
G9	command-r7b-12-2024	0.900	0.897	0.690	0.929

E2-A/E2-B union: NOT_COMPUTED (E2-A missing). Code freeze: 78 files checked, 0 mismatches. Exp-5 reproduction: c_score_align 50/50, p_peer_text 50/50. Score seal: 77 files hashed, 0 access violations.

6.4 Experiment 13: R_COMP FREE Confirmation

Gloss gate: FAILED twice. Run 1: haiku BA 0.776, qwen 0.855, both_yes 0.817—all below 0.90. Run 2: haiku BA 0.561, qwen 0.883, both_yes 0.765—failed again.

Confirmatory verdict for criterion (c): NOT_TESTABLE. Zero CORRECT rows exist.

6.4.1 Provisional results (ERROR_CERT vs MAPPED, non-confirmatory)

Table 79: R_COMP FREE provisional AUROCs (within-template). Iteration 5, Experiment 13.

Metric	Within-template AUROC [CI]
c_score_align	0.801 [0.770, 0.834]
c_score_hyb	0.817 [0.788, 0.848]
c_score_nf	0.743 [0.710, 0.776]
c_exact	0.606 [0.580, 0.633]
judge_cheap_disg	0.584 [0.550, 0.618]
judge_cheap_orig	0.633 [0.601, 0.666]

c.align minus judge_cheap_disg: $\Delta = +0.217 [+0.175, +0.260]$ ($p < 0.0001$). c_hyb is the highest provisional AUROC at 0.817.

6.4.2 Blind audit

Table 80: Blind two-family audit (100 per class, untouched). Iteration 5, Experiment 13.

Auditor	ERROR_CERT precision	MAPPED faithful
GLM-4.6	0.720 [0.622, 0.801]	0.904 [0.828, 0.949]
Sonnet-5	0.845 [0.760, 0.904]	0.722 [0.625, 0.801]
Consensus (both agree)	0.838 [0.738, 0.905]	0.917 [0.830, 0.961]

6.5 Experiment 14: Freeze and GG Screening

Workspace: gen_art_experiment_14.

V0 (c_score_align) is frozen. Five variants (V1–V5) screened against it on development set E ($Z = 2,672$ R_AB rows: 1,810 ERROR / 862 CORRECT, 283 sentences, ≥ 2 cross-family peers). CIs: sentence-cluster percentile bootstrap, $B = 2000$, seed 20260924.

6.5.1 T1. Reproduction gates

6.5.2 T2. AUROC per variant and cell

Cell sizes (n /ERROR/CORRECT/sentences): ALL-strat 2672/1810/862/283; LONG-strat 2289/1664/625/247; L25 869/694/175/108; L20 818/589/229/77; EXC 602/381/221/62; CTRL 383/146/237/36.

Table 81: Reproduction gates (development E). `gen_art_experiment_14`.

Gate	Reproduced	Target	Pass
G0 V0 strat AUROC (R_AB 2,686)	0.7415	0.7415	True
G0 Δ (V0 – judge disg) strat	0.0990 [0.049, 0.146]	+0.099	True
G0 Δ long pool	0.1163	+0.116	True
G1 V0.rep vs frozen ($ \text{diff} > 10^{-6}$)	24/8469 = 0.28%	$\leq 0.5\%$	True
c.exact pooled AUROC on Z	0.7477	0.748	True
3-pool plain c pooled AUROC	0.7845	0.7845	True
eval-3 d.oracle 3-pool ALL / L25	0.3854 / 0.5422	0.385 / 0.542	True
eval-3 END_MAJ 3-pool L25 d	0.7349	0.735	True

Table 82: Strat AUROC per variant and cell (Z, development E). `gen_art_experiment_14`.

Score	ALL-strat	pooled	LONG-strat	L25	L20	EXC	CTRL
V0 frozen	0.742	0.784	0.743	0.713	0.724	0.815	0.738
V0 re-deriv.	0.742	0.783	0.742	0.712	0.724	0.815	0.738
c.exact	0.689	0.748	0.672	0.611	0.652	0.794	0.857
V1 plur.-norm.	0.729	0.766	0.739	0.704	0.727	0.809	0.628
V2 rel.-wt.	0.752	0.790	0.750	0.718	0.737	0.819	0.773
V3 = V1+V2	0.737	0.774	0.750	0.714	0.743	0.812	0.613
V4 two-ch.	0.758	0.799	0.751	0.719	0.732	0.830	0.826
V5 = V3/3p	0.735	0.775	0.746	0.694	0.771	0.780	0.629
judge disg	0.641	0.696	0.624	0.641	0.615	0.615	0.807
judge orig	0.624	0.636	0.629	0.653	0.611	0.624	0.571
judge2 orig	0.653	0.701	0.644	0.671	0.631	0.624	0.740

Table 83: Paired Δ strat AUROC vs V0 (frozen), same rows. `gen_art_experiment_14`.

Variant	ALL-strat	pooled	LONG	L25	L20	EXC	CTRL
V0 re-deriv.	−0.000	−0.000	−0.000	−0.001	0.000	0.000	−0.000
c.exact	−0.053	−0.036	−0.070	−0.102	−0.072	−0.022	+0.119
V1	−0.013	−0.017	−0.004	−0.009	+0.003	−0.006	−0.110
V2	+0.010	+0.006	+0.008	+0.005	+0.013	+0.003	+0.036
V3	−0.005	−0.010	+0.007	+0.001	+0.020	−0.003	−0.125
V4	+0.016	+0.016	+0.009	+0.006	+0.008	+0.015	+0.088
V5	−0.007	−0.009	+0.003	−0.019	+0.047	−0.036	−0.109

6.5.3 T3. Paired Δ AUROC vs V0 (frozen)

6.5.4 T4. Pre-registered selection

Decision: NONE—V0 stands. Rule: winner = variant with highest E LONG-strat AUROC (Z) that (i) beats V0 by $\geq +0.015$ there AND (ii) $\geq$ V0 on L25 (point) AND (iii) $\geq$ V0 -0.01 on CTRL; none $\rightarrow$ “NONE: V0 stands”.

Table 84: Pre-registered selection (5 variants screened). `gen_art_experiment_14`.

Variant	Δ LONG	(i) $\geq +0.015$	(ii) L25 $\geq$ V0	(iii) CTRL	Qualifies
V1	−0.004	False	False	False	False
V2	+0.008	False	True	True	False
V3	+0.007	False	True	False	False
V4	+0.009	False	True	True	False
V5	+0.003	False	False	False	False

Selection stability ($B = 500$): NONE: V0 stands 0.876, V4 0.118, V5 0.004, V3 0.002. Permutation null ($B = 200$): share of permutations in which ANY variant qualifies = 0.035 (V5 only).

6.5.5 T5. Binary decomposition (e/d at $c > 0.5$) and NET

Table 85: e / d at $c > 0.5$ (V4 flag rate matched to V0). `gen_art_experiment_14`.

Score	ALL e/d	LONG	L25	CTRL	words T1	words T2	words T3
V0 frozen	.140/.490	.094/.651	.050/.811	.671/.063	.311/.254	.082/.716	.053/.913
V0 re-deriv.	.141/.490	.094/.651	.050/.811	.678/.063	.313/.254	.082/.716	.053/.913
c.exact	.018/.722	.008/.918	.003/.994	.123/.203	.043/.502	.010/.982	.004/1.00
V1	.343/.174	.309/.224	.241/.417	.733/.042	.463/.061	.355/.225	.216/.513
V2	.137/.483	.091/.642	.045/.811	.658/.063	.307/.254	.081/.695	.047/.913
V3	.303/.186	.266/.240	.197/.434	.726/.042	.438/.066	.296/.251	.185/.522
V4	.126/.457	.079/.606	.033/.834	.664/.063	.287/.201	.076/.687	.037/.957
V5	.240/.240	.202/.312	.150/.509	.678/.051	.371/.114	.219/.316	.144/.574

V0 flag rate on Z : 0.7403; V4 matched threshold: 0.6272.

Table 86: NET $\Delta(e + d)$, words T3 – T1 (eval-2 definition). `gen_art_experiment_14`.

Score	NET [CI]	Δe	Δd
V0 frozen	+0.401 [0.259, 0.533]	−0.258	+0.659
V0 re-deriv.	+0.399 [0.257, 0.532]	−0.260	+0.659
c.exact	+0.458 [0.323, 0.579]	−0.040	+0.498
V1	+0.205 [0.067, 0.352]	−0.246	+0.452
V2	+0.399 [0.260, 0.535]	−0.260	+0.659
V3	+0.202 [0.069, 0.344]	−0.254	+0.456
V4	+0.506 [0.378, 0.624]	−0.250	+0.755
V5	+0.233 [0.087, 0.390]	−0.227	+0.460

6.5.6 T5b. Key diagnostic: does d fall without e rising?

No variant achieves “ d falls without e rising” in any cut. Selected rows (the full table of 35 rows covering all variants $\times$ cuts is in the artifact workspace):

Table 87: Key diagnostic: Δd and Δe vs V0 at selected cuts. `gen_art_experiment_14`.

Variant	Cut	Δd [CI]	Δe [CI]	d falls w/o e rise
c_exact	words T3	+0.087 [0.016, 0.182]	−0.049 [−0.097, −0.012]	False
V1	words T3	−0.400 [−0.500, −0.283]	+0.163 [0.124, 0.202]	False
V2	words T3	0.000 [0.000, 0.000]	−0.005 [−0.018, 0.004]	False
V3	words T3	−0.391 [−0.486, −0.275]	+0.132 [0.098, 0.166]	False
V4	words T3	+0.043 [0.000, 0.111]	−0.016 [−0.042, 0.006]	False
V5	words T3	−0.339 [−0.453, −0.212]	+0.091 [0.059, 0.123]	False
c_exact	L25	+0.183 [0.074, 0.304]	−0.048 [−0.086, −0.015]	False
V1	L25	−0.394 [−0.503, −0.283]	+0.190 [0.152, 0.229]	False
V4	L25	+0.023 [−0.078, 0.125]	−0.017 [−0.041, 0.004]	False
c_exact	ALL	+0.232 [0.174, 0.294]	−0.123 [−0.164, −0.088]	False
V1	ALL	−0.316 [−0.383, −0.256]	+0.203 [0.175, 0.231]	False
V4	ALL	−0.032 [−0.073, 0.008]	−0.014 [−0.028, 0.000]	False

6.5.7 T6. Error-endorsement e per T9 error class

Table 88: e per T9 error class (ERROR rows not flagged). `gen_art_experiment_14`.

Class	n	V0	V0 re-d.	c_exact	V1	V2	V4	V5
ADD	1033	0.151	0.151	0.026	0.328	0.142	0.131	0.229
DROP	1395	0.151	0.151	0.019	0.337	0.147	0.129	0.249
COMPOUND	1141	0.032	0.032	0.006	0.194	0.029	0.030	0.119
polarity	432	0.056	0.056	0.000	0.190	0.053	0.037	0.111
structural	1293	0.082	0.083	0.004	0.275	0.081	0.073	0.172
MR-type	219	0.502	0.507	0.009	0.872	0.525	0.411	0.721

6.5.8 T7. Oracle bound (eval-3, label-using diagnostic)

Table 89: Oracle bound (d_{oracle} , label-using); selected rows. `gen_art_experiment_14`.

Pool	Cut	d_{oracle}	d_{V0}	Variant	d_v	e_v	gap_closed
9fam	ALL	0.586	0.490	V0	0.490	0.140	ILL_COND
9fam	LONG	0.758	0.651	V0	0.651	0.094	ILL_COND
9fam	L25	0.834	0.811	V0	0.811	0.050	ILL_COND
9fam	L25	0.834	0.811	V4	0.834	0.033	ILL_COND
9fam	words T3	0.887	0.913	V0	0.913	0.053	ILL_COND
9fam	words T3	0.887	0.913	V4	0.957	0.037	ILL_COND
3pool	ALL	0.385	0.486	V5	0.230	0.246	+2.535
3pool	LONG	0.488	0.649	V5	0.301	0.207	+2.162
3pool	L25	0.542	0.813	V5	0.482	0.159	+1.222
3pool	words T3	0.624	0.917	V5	0.550	0.154	+1.250

The full oracle table (24 rows covering all variants $\times$ pools $\times$ cuts) is in the artifact workspace.

6.5.9 T8. Context: Δ vs cheap API judges

The full table (18 contrasts: V0/V1/V2/V3/V4/V5 $\times$ 3 judges) is in the artifact workspace.

Table 90: Δ vs cheap API judges (frozen exp-6 columns); selected rows. `gen_art_experiment_14`.

Contrast	ALL-strat		LONG-strat		L25	
V0 – judge disg	+0.101	[0.053, 0.145]	+0.118	[0.068, 0.163]	+0.072	[−0.015, 0.153]
V0 – judge orig	+0.119	[0.079, 0.157]	+0.114	[0.073, 0.153]	+0.060	[−0.003, 0.125]
V0 – judge2 orig	+0.090	[0.044, 0.136]	+0.099	[0.053, 0.147]	+0.042	[−0.046, 0.129]
V4 – judge disg	+0.117	[0.069, 0.162]	+0.127	[0.077, 0.174]	+0.077	[−0.011, 0.160]
V4 – judge orig	+0.135	[0.093, 0.175]	+0.122	[0.081, 0.164]	+0.066	[−0.006, 0.135]
V4 – judge2 orig	+0.106	[0.060, 0.154]	+0.108	[0.060, 0.159]	+0.047	[−0.049, 0.136]

6.5.10 T9–T10. GG checker gates and development gates

Table 91: GG checker gates (Gemini 2.5 Flash, non-thinking, $T = 0$). `gen_art_experiment_14`.

Gate	Half	Bal. acc.	TPR	TNR	n YES/NO
G-A	dev	0.780	0.789	0.770	190/252
G-B	dev	0.950	0.966	0.934	29/91
G-A	confirm	0.907	0.841	0.974	170/228
G-B	confirm	0.941	0.917	0.965	24/85

G-B has 53 YES / 176 NO pairs (< 60 YES): NOT_TESTABLE as a gate.

Table 92: GG development gates $\rightarrow$ status FAIL(g3). `gen_art_experiment_14`.

Gate	Value	Rule	Pass
g1 SYN flip	0.034	$\text{flip} \leq 0.05 \ \& \ \text{FA} \leq \text{base} + 0.05$	True
g2 MR recall	GG 1.000 vs c.align 1.000	$\geq \text{c.align} - 0.05$	True
g3 E strat AUROC	GG 0.696 vs V0 0.742	$\geq \text{V0} - 0.01$	False
checker	G-A confirm BA 0.907	$\text{G-A} \geq 0.90$	True

RENAME_NONCE (pre-stated expected failure, not gated): FA 1.000, paired-base FA 0.773, paired flip 0.227 ($n = 110$).

6.5.11 T10b. Rename invariance per metric and control type

Table 93: Rename invariance / MEANING_RENAME recall per metric (PERTURB E-bases, $c > 0.5$). `gen_art_experiment_14`.

Metric	Base FA	SYN FA / flip	NONCE FA / flip	MR recall	MR base endorsed
c-gg	0.828	0.932 / 0.034	1.000 / 0.227	1.000	DOWN 1.0, UP 1.0
c-gg_allyes	0.798	0.864 / 0.000	0.745 / 0.000	0.821	DOWN 0.84, UP 0.80
c_exact_pt	0.894	1.000 / 0.068	1.000 / 0.136	1.000	DOWN 1.0, UP 1.0
c_align	0.712	0.943 / 0.182	1.000 / 0.327	1.000	DOWN 1.0, UP 1.0

6.5.12 T11. GG on E (Z)

Non-exact E node pairs: 11,889; gloss outcomes: PREFILTER 10,514, gloss_rejected 403, gloss_yes 675, identity_map 12, NO_MAP 285. Map found: 9.2%; gloss-rejected: 3.4%. The 20 gloss-

Table 94: GG AUROC on development E (Z). `gen_art_experiment_14`.

Score	ALL-strat	pooled	LONG-strat	L25	CTRL
GG (PRIMARY)	0.696	0.751	0.689	0.657	0.773
GG all-yes	0.685	0.745	0.683	0.652	0.709
c_exact	0.689	0.748	0.672	0.611	0.857
V0 frozen	0.742	0.784	0.743	0.713	0.738
Δ GG – V0	−0.046	−0.033	−0.054	−0.056	+0.035

Table 95: GG e/d and per-class e on E. `gen_art_experiment_14`.

Score	ALL e/d	LONG e/d	L25 e/d	words T3 e/d
GG (PRIMARY)	0.081/0.629	0.038/0.822	0.022/0.960	0.026/0.983
GG all-yes	0.113/0.616	0.069/0.808	0.048/0.943	0.047/0.974
c_exact	0.018/0.722	0.008/0.918	0.003/0.994	0.004/1.000
V0 frozen	0.140/0.490	0.094/0.651	0.050/0.811	0.053/0.913

Table 96: T9 e by error class, GG vs V0 on E. `gen_art_experiment_14`.

T9 class	n	GG e	GG all-yes e	exact e	V0 e
ADD	1033	0.085	0.118	0.026	0.151
DROP	1395	0.081	0.116	0.019	0.151
COMPOUND	1141	0.018	0.030	0.006	0.032
polarity	432	0.023	0.030	0.000	0.056
structural	1293	0.039	0.062	0.004	0.082
MR-type	219	0.224	0.338	0.009	0.502

disagreement examples (GG YES vs eqmv rejected, and vice versa) are in the artifact workspace (`tables.md T11c`).

6.5.13 T12. GG cost

Table 97: GG cost (development E + PERTURB). `gen_art_experiment_14`.

Item	Value
Gloss sweep \$ (E + PERTURB)	\$0.1630
\$ per call (≤ 12 pairs)	\$0.000393
FULL per E candidate: CPU s / \$ (upper)	0.038 s / \$0.000047
MARGINAL per candidate: CPU s / \$	0.106 s / \$0.000010
Peer generation \$ per sentence (7 fam., shared with V0)	\$0.002206

6.5.14 T13. SECONDARY GG_B (PRIMARY-accepted pairs kept)

Status: UNTESTED(budget): the run-level budget was exhausted before the GG_B gloss (HTTP 403); GG_B agreement is bracketed by `c_gg` (no bridged map accepted) and `c_ggb_allyes` (every bridged map accepted). Agreement sets are nested ($GG \subseteq GG_B \subseteq GG_B \text{ all-yes}$).

Table 98: GG_B AUROC brackets on E. `gen_art_experiment_14`.

Score	ALL-strat	pooled	LONG-strat	L25	CTRL
GG_B all-yes	0.653	0.697	0.669	0.631	0.495
GG (PRIMARY)	0.696	0.751	0.689	0.657	0.773
<code>c_exact</code>	0.689	0.748	0.672	0.611	0.857
V0 frozen	0.742	0.784	0.743	0.713	0.738
Δ <code>c_ggb_allyes</code> – V0	−0.090	−0.087	−0.074	−0.082	−0.243
Δ <code>c_gg</code> – V0	−0.046	−0.033	−0.054	−0.056	+0.035

Table 99: GG_B e/d brackets and per-class e . `gen_art_experiment_14`.

Score	ALL e/d	LONG e/d	L25 e/d	words T3 e/d
GG_B all-yes	0.334/0.358	0.294/0.472	0.264/0.623	0.246/0.739
GG (PRIMARY)	0.081/0.629	0.038/0.822	0.022/0.960	0.026/0.983
<code>c_exact</code>	0.018/0.722	0.008/0.918	0.003/0.994	0.004/1.000
V0 frozen	0.140/0.490	0.094/0.651	0.050/0.811	0.053/0.913

GG_B search: 11,202 pairs; status MAPPED 3,133, NO_MAP 7,899, CAPPED 170; CPU 3,326s; bridges in accepted maps: B2 1,983, B1 2,616, B4 1,470, B3 367, B5 34. Gloss: 11,956 items, 1,090 calls, all NOT_RUN(QuotaError).

6.5.15 T14. Independent audit

Decision re-derived: NONE: V0 stands (match True); best LONG Δ V4 +0.00880 (match 10^{-9} True); GG gate numbers match: True; shuffled-label placebo strat AUROCs (V0 0.498, V1 0.497, V2 0.497, V3 0.495, V5 0.498, V4 0.494, `c_gg` 0.506): all near 0.5 (ok True).

6.6 Evaluation 4: Corrected Record

Workspace: `gen_art_evaluation_4`.

244 claims verified against source files; hypothesis text mismatching its source: 0. **Paper-vs-file mismatches: 15**—the values the iteration-4 paper prints differ from the source file. Key corrections: E is development data not held-out; T2 NOT CONFIRMED (corrected from “T2 CONFIRMED”); MEANING_RENAME is an ERROR operator; cost units corrected; mechanism verdicts aligned with pre-registered definitions.

Table 100: Paper-vs-file mismatches found by `gen_art_evaluation_4` (15 values).

Claim	Quantity	Paper prints	File value
x9_csc	c_csc strat AUROC CI	[0.545, 0.679]	0.614 [0.493, 0.736]
x9_fexact	FREE_exact strat CI	[0.704, 0.827]	0.770 [0.686, 0.844]
x9_calign	c_score_align strat CI	[0.711, 0.827]	0.774 [0.682, 0.857]
x9_ppt	p_peer_text strat CI	[0.742, 0.848]	0.799 [0.700, 0.880]
x9_s4	S4_full strat CI	[0.672, 0.791]	0.736 [0.635, 0.826]
x9_flash	flash-lite disg strat CI	[0.592, 0.731]	0.664 [0.568, 0.750]
x9_t9_COMP_CSC	e_CSC COMPOUND	0.643	0.282
x9_t9_COMP_exact	e_exact COMPOUND	0.246	0.107
x9_g2	G2 verdict	FAIL	NOT_READ
x9_m3.boot	M3 bootstrap CI	[−0.417, −0.041]	−0.232 [−0.770, −0.023]
rv_spend3	iter-3 spend	\$1.58	\$4.33
rv_k3_cost	k=3 \$/sentence	\$0.000073	\$0.000946
rv_frontier_x	frontier/consensus ratio	6.07×	30.2×
rv_k1_cost	k=1 \$/sentence	\$0.000024	\$0.000315
rv_k7_cost	k=7 \$/sentence	\$0.000170	\$0.002206

6.6.1 Corrected rename invariance (PERTURB, development data)

Table 101: Invariance on meaning-preserving renames of CORRECT E bases (FA = share flagged, $c > 0.5$; development). `gen_art_evaluation_4`.

Metric	NONCE FA [CI]	NONCE base FA	NONCE flip	SYN FA [CI]	SYN base FA	SYN flip
c_align	0.765	0.160	0.605	0.565 [0.487, 0.634]	0.252	0.313
c_nf	0.132 [0.090, 0.175]	0.132	0.000	0.201 [0.148, 0.254]	0.201	0.000
c_hyb	0.216 [0.146, 0.285]	0.139	0.077	0.298 [0.212, 0.385]	0.270	0.029
judge disg	0.229 [0.194, 0.264]	0.109	0.134	0.085 [0.054, 0.121]	0.099	0.058
p_peer_text	1.000	0.179	0.821	0.216 [0.136, 0.307]	0.136	0.080
p_text	1.000	0.109	0.891	0.068 [0.023, 0.125]	0.068	0.000
l3_z3	0.385 [0.296, 0.473]	0.109	0.276	0.023 [0.000, 0.057]	0.023	0.000
S4_local	0.804 [0.732, 0.875]	0.241	0.616	0.364 [0.273, 0.466]	0.193	0.239

Pre-registered rename-invariant selection FAILED: HYB PERTURB RENAME_SYN / NONCE FA 0.841 / 0.700; NF-anchored 0.686 / 0.485; both INELIGIBLE, nothing was frozen.

Table 102: Mechanism verdicts under pre-registered definitions (development E). `gen_art_evaluation_4`.

Clause	Verdict	Key number
M1 (endorsement vs conditions)	INCONCLUSIVE	
M2 (divergence vs words)	CONFIRMED, non-specific	shuffled placebo also passes
M3 (specifications conflict)	INCONCLUSIVE	boot +0.146 [−0.051, 0.361]
M4 ($k_{95} \leq 5$ families)	CONFIRMED	$k_{95} = 3$ (T3: 5)
NET $\Delta(e + d)$ T3–T1	degrades	+0.356 [0.198, 0.493]

Table 103: Corrected k -saturation (FULL \$ per sentence, peer generation). `gen_art_evaluation_4`.

k	AUROC	AUROC words T3	\$ per sentence (FULL)
1	0.685	0.581	\$0.000315
3	0.757	0.651	\$0.000946
5	0.776	0.698	\$0.001576
7	0.784	0.726	\$0.002206

Table 104: Machine-verified corrections to iterations 1–2. `gen_art_evaluation_4`.

ID	Claim as reported	Corrected value	Verified
C1a	iter-2: contamination DiD on E not reported for exp 6	Llama-8B DiD +0.162 [0.064, 0.261]	True
C1b	same	Qwen-8B DiD +0.036 [−0.039, 0.111]	True
C1c	exp 5: disguise IMPROVES Qwen judge on E	n.s. −0.064 [−0.098, −0.032]	True
C2_SOLVER	disguise “forces structural reading”	Orig ahead: +0.106 [0.033, 0.178]	True
C2_ADJ	same (R_ADJ_AB)	Orig ahead: +0.198 [0.130, 0.260]	True
C3	fusion OOF 0.943 as quality	shrinks to held-out E R_AB 0.790	True
C4	frontier-judge advantage weakens	STRENGTHENS: +0.077 → +0.166	True
C5	P1 group names mixed	local judge recall at FA 0.10: 0.169 / 0.199	True
C6	P4 FA 0.385 vs “0.013”	0.013 was FOL-Triage FLIP rate	True
C7	iter-2 spend statement	total \$4.229	True

Table 105: Hypothesis verdicts, iteration 3 (corrected). `gen_art_evaluation_4`.

Clause	Verdict	Reason
T1(a)	CONFIRMED on dev E; L25 n.s.	L25 +0.069 [−0.020, 0.148]
T1(b)	CONFIRMED (nested only)	
T1(c)	NOT CONFIRMED	SIG PASS / FREE NOT_TESTABLE
T1(d)	point estimate only	ratio 0.957 [0.887, 1.034]
T1(e) rename	FAILED	
M1	INCONCLUSIVE	
M2	CONFIRMED, non-specific	shuffled placebo passes
M3	INCONCLUSIVE	specifications conflict
M4	CONFIRMED	
NET	degrades	
Typing	at judge level (negative)	

Table 106: Hypothesis verdicts, iteration 4 (PROVISIONAL where stated). `gen_art_evaluation.4`.

Clause	Verdict	Source
CSC G1 ($d \leq 0.35$ AND slope)	FAIL: d 0.139, slope +1.25 [0.41, 2.09]	csc_gate_E.json
CSC G2 (strat AUROC $\geq V0 - 0.01$)	NOT_READ (R_AB part FAIL)	csc_gate_E.json
CSC G3-E (RENAME_SYN FA gate)	NOT_RUN	csc_gate_E.json
CSC G4	null on E (sibling arm)	csc_gate_E.json
CSC G5 (FULL \$/candidate $\leq \$0.002$)	PASS (PROV.): $\$5.25e-4$	csc_gate_E.json
exp-10 G3-P / G4 / MT	UNTESTED (0 CSC peers; budget)	csc_gate_P.json
E2 confirmation cells	NOT TESTABLE (0 CORRECT rows)	testability_E2.json
R_COMP FREE criterion (c)	NOT TESTABLE (0 CORRECT; no gloss)	testability_FREE_v2.json

Table 107: Cost table with corrected units. `gen_art_evaluation.4`.

Item	Unit	FULL/MARG	\$	Ratio vs flash-lite
judge.cheap	per item-call	FULL (API)	$\$4.88e-5$	$1.00\times$
judge.cheap2	per item-call	FULL (API)	$\$2.01e-5$	$0.41\times$
strong	per item-call	FULL (API)	$\$3.65e-3$	$74.91\times$
roundtrip	per item-call	FULL (API)	$\$2.75e-5$	$0.56\times$
sc5	per item-call	FULL (API)	$\$6.65e-5$	$1.36\times$
consensus c_score.align	per candidate	FULL (gen+z3)	$\$1.21e-4$	$2.48\times$
k-pool $k = 1$	per sentence	FULL (gen)	$\$3.15e-4$	—
k-pool $k = 3$	per sentence	FULL (gen)	$\$9.46e-4$	—
k-pool $k = 5$	per sentence	FULL (gen)	$\$1.58e-3$	—
k-pool $k = 7$	per sentence	FULL (gen)	$\$2.21e-3$	—
SIG peer generation	per candidate	FULL (gen)	$\$1.16e-4$	$2.39\times$
SIG peer generation	per sentence (10 slots)	FULL (gen)	$\$1.16e-3$	—
marginal z3 check	per candidate	MARGINAL (z3)	$\$2.50e-7$	$0.01\times$
CSC (exp 9, G5)	per candidate	FULL (undeduped)	$\$5.25e-4$	$10.77\times$
CSC (exp 9, G5)	per candidate	FULL (dedup)	$\$2.76e-4$	$5.67\times$
E2 pilot, L25	per sentence	FULL (projection)	$\$0.0118$	—

6.6.2 Corrected mechanism verdicts (pre-registered definitions)

6.6.3 Iteration-1/2 corrections (11 items)

6.6.4 Hypothesis verdicts (iteration 3)

6.6.5 Hypothesis verdicts (iteration 4, development data)

6.6.6 Cost table (units explicit)

The frontier judge costs $30.2\times$ the consensus score per item (FULL); the ratio in the iteration-3 paper mixed a per-call judge price with a per-candidate generation price.

6.6.7 Iteration-4 spend (from ledgers)

Table 108: Iteration-4 spend by artifact (ledger window). `gen_art_evaluation_4`.

Artifact (ledger dir)	\$ in window	Records
iter_3/gen_art_experiment_6	\$2.3953	36926
iter_3/gen_art_experiment_7	\$1.5974	11389
iter_3/gen_art_experiment_8	\$0.3452	6710
iter_4/gen_art_dataset_4	\$0.2409	412
iter_4/gen_art_experiment_9	\$0.1082	1166
iter_4/gen_art_experiment_10	\$0.0000	2
iter_3/gen_art_research_1	\$0.0000	8

Iteration-4 artifacts: \$0.349 (exp 9 \$0.108, exp 10 \$0.0000088, dataset 4 \$0.241, dataset 5 \$0; eval 3 has no ledger, CPU only). Other artifacts whose ledger records fall in the same window: \$4.338 (iteration-3 experiments). Evidenced total: \$4.687 vs the \$7.00 Test-idea budget: unaccounted \$2.313 (no ledger in the run tree records it).

6.6.8 Dead ends (extended)

Table 109: Dead ends accumulated across all iterations. `gen_art_evaluation_4`.

Dead end	Iter	Reason
Rename-invariant selection (NF-anchored and HYB)	3	both INELIGIBLE (FA-only selection rule, no flip): SYN/NONCE FA NF 0.686/0.485, HYB 0.841/0.700; nothing frozen
Name-free consensus c_nf	2-3	pre-registered selection failure + MEANING.RENAME blindness (0.499)
Candidate B1	1	planned, never ran (workspace holds 0 files — empty)
FOL-Triage fused score	1	fails the rewrite-FA gate (worst-family FA 0.275 vs 0.10 bar)
P2 / P4 (exp 5)	2	P2 REFUTED, P4 REFUTED
B2 decomposed z3 reader	2	gate failure: balanced accuracy 0.476
R-ADJ LLM adjudicator	2	gate failure: Sonnet-5 BA 0.767 overall, 0.662 on expert track-H
CSC, CSC_graded, CSC-multi, HYB_MEAN	4	all fail G1 (d slopes positive); CSC provisional negative on completion-selected subset; L25 and RENAME untested
E2 EXC-core stratum	4	core-exception supply exhausted (D3)
CSC arms and panel route for R.COMP FREE	4→5	CLOSED by the iteration-5 hypothesis
Vocabulary-divergence diagnosis of iteration 3	3→4	refuted on E by label-based decomposition

6.7 Iteration 5 Summary

Confirmation is NOT_TESTABLE across the board. E2-A: flash-lite bar has zero scored rows. E2-B: 48 CORRECT < 50. R_COMP FREE: gloss gate failed twice, zero CORRECT rows. The run-wide budget (\$7.00) was exhausted before any confirmation arm could complete.

DT stratum is a bright spot. V0 reaches strat AUROC 0.916 [0.858, 0.970] on 179 ProverQA-derived rows with trusted gold.

L25 on E2 is near chance. V0 strat AUROC 0.443 on E2-A L25 (42 rows, 7 CORRECT, 6 sentences).

V0 stands. No variant V1–V5 met the selection rule. GG fails its development gate.

Rename weakness persists. V0 RENAME_NONCE FA 1.0 on E2; RENAME_SYN FA 0.944. Figure 8 summarizes the vocabulary-source tradeoff across all iterations.

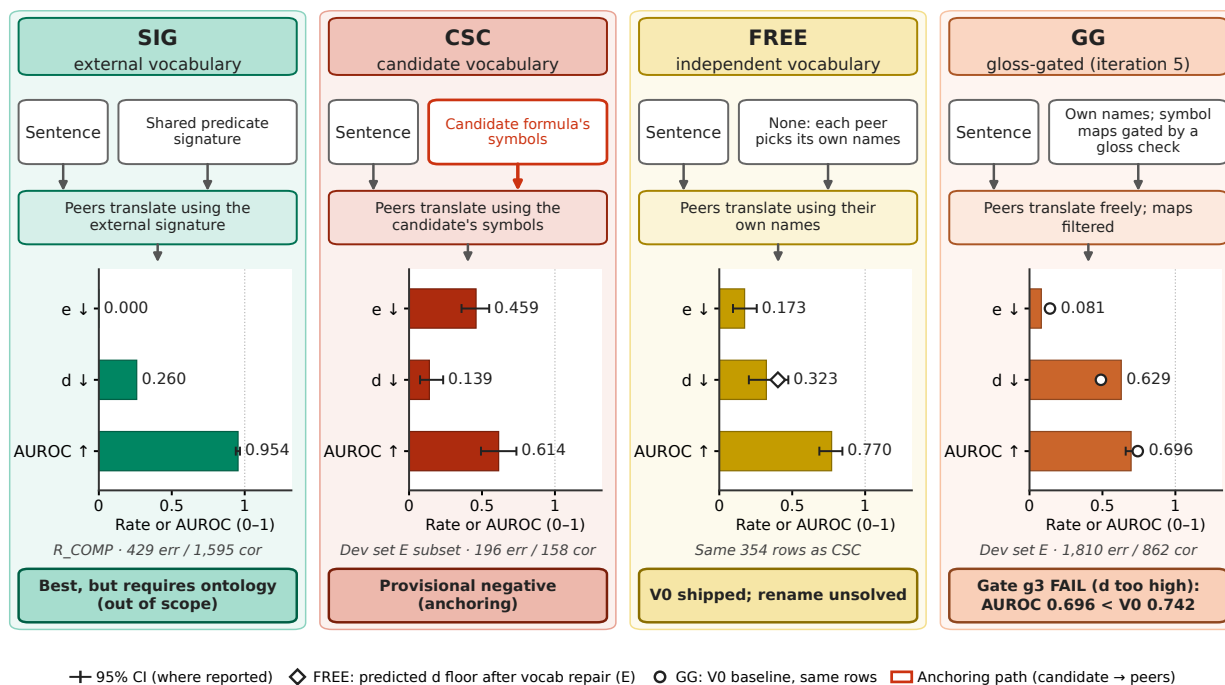


Figure 8: The vocabulary-source tradeoff: three vocabulary regimes for cross-family consensus and their effects on error endorsement (e), correct-row divergence (d), and AUROC. External vocabulary (SIG) achieves near-perfect discrimination; candidate vocabulary (CSC) introduces anchoring; independent vocabulary (FREE) preserves independence but raises d . GG (iteration 5) attempted to filter FREE vocabulary mappings but failed its development gate.

Review score: 3 (blocking). Hypothesis update: move DECLARE. Title: “Cross-model agreement, unconfirmed on fresh data.” Iteration-5 API spend: E2-B \$2.05; R_COMP FREE \$1.97; other artifacts CPU-only.

7 What the Run Learned

The cross-family consensus metric (c_score_align) is the strongest gold-free faithfulness signal found in this investigation. On development set E, it achieves stratified AUROC 0.742, beating the best cheap LLM judge (0.641) by +0.101 [+0.053, +0.145]. On R_COMP SIG (the out-of-scope

Table 110: Dead ends across all iterations (final). Iteration 5.

Dead end	Iter.	Reason
L1 lint on LLM outputs	1	AUROC 0.508
L2 role-aware	1	AUROC 0.557
Pilot structural metrics	1	AUROC 0.500–0.533 (null)
Round-trip reformalisation	1	AUROC 0.513 (76% ties)
B2 decomposed z3 reader	2	Gate balanced accuracy 0.476
R_ADJ LLM adjudicator	2	Sonnet-5 balanced accuracy 0.662
Name-free consensus (c_nf)	2–3	SWAP recall 0.512
CSC (provisional negative)	4	Strat AUROC 0.614, anchoring $e = 0.459$
GG (Gloss-Gated agreement)	5	Gate g3 FAIL: strat AUROC 0.696 vs V0 0.742

controlled-vocabulary condition), it reaches 0.954 within-template AUROC with zero error endorsement. Three peer families suffice for 97% of full-pool performance (M4 confirmed: $k_{95} = 3$; by words tercile $k_{95} = 2/3/5$).

The mechanism is clear: errors scatter across z3-equivalence classes while correct translations cluster. The consensus signal degrades with sentence length (NET +0.401 on development E, driven by divergence d), meaning longer sentences need more peers and the metric becomes less reliable precisely where it matters most.

Iteration 5 attempted confirmation on three fronts, and all three were blocked by the run-wide budget (\$7.00). E2-A produced one testable cell: V0 reaches 0.916 on ProverQA-derived sentences with trusted gold, but only 0.443 on the L25 long-sentence stratum (near chance). E2-B’s exploratory results (V0 0.737 vs flash-lite 0.496, $\Delta + 0.208$) favour consensus on long sentences with unaudited labels, but the label-coupling diagnostic shows part of the advantage comes from shared gold-equivalent peers. R_COMP FREE’s provisional results show c_align 0.801 vs judge 0.584 ($\Delta + 0.217$) under ERROR_CERT vs MAPPED labels, with consensus carrying most of the signal in nesting tests.

No consensus variant V1–V5 improved on V0. The best candidate (V4, two-channel logistic) gained only +0.0088 LONG-strat, well below the +0.015 selection threshold. V0 stands as the shipped metric.

The rename tradeoff remains the central open problem. V0 has RENAME_NONCE FA 0.765 on development E and 1.0 on E2: any renaming of predicates breaks the consensus signal. The GG checker reduced MEANING_RENAME endorsement from 0.502 (V0) to 0.224, but failed the development gate (strat AUROC 0.696 vs V0 0.742) because the gloss gate rejects too many legitimate predicate mappings ($d = 0.629$).

Five limitations bound the current results:

1. No confirmatory hypothesis is confirmed or refuted: every confirmation arm was budget-stopped before reaching testability.
2. All evaluation is on FOLIO and MALLS-derived data; domain transfer is untested.
3. The labels are noisy: the panel majority accuracy on expert-curated errors is 0.727, and the R_COMP FREE audit shows 13–28% label contamination.
4. The consensus advantage does not grow with sentence complexity (M3 INCONCLUSIVE).
5. The rename tradeoff is unsolved: no variant achieves both low rename FA and high error recall.

8 Run Bookkeeping

Table 111: Run trajectory: one row per iteration. Ledger spend is cumulative LLM/tool/container spend (excludes compute-rental time).

Iter.	Hypothesis title	Move	Review	Executed	Ledger spend
1	Peer agreement plus text checks vs judges	deepen	3	yes	\$17.37
2	Peer agreement vs judges on long sentences	deepen	3	yes	\$34.76
3	Peer agreement in the candidate’s own words	deepen	3	yes	\$84.75
4	Cross-model agreement, tested on fresh data	deepen	3	yes	\$159.80
5	Cross-model agreement, unconfirmed on fresh data	declare	3	yes	\$315.97

9 Related Work

NL-FOL datasets and their quality. FOLIO provides 1,435 FOL-annotated NLI problems. MALLS uses GPT-4 to generate and verify NL-FOL pairs at scale. Brunello et al. found that approximately 42% of entries in both datasets contain incorrect FOL formalisations. Logic-LM prompts LLMs for FOL and calls a prover, releasing outputs that we use as our evaluation screen.

FOL closeness metrics. Thatikonda et al. evaluate n-gram, graph, embedding, and LLM-based metrics on controlled FOL perturbations, finding that no single metric is sensitive to all perturbation types.

Factuality metrics. FRANK defines a typology of factuality errors in abstractive summarisation. QuestEval uses question generation and answering to assess factual consistency. QAGS generates questions from a summary and checks whether the source answers them the same way. Our L3 questionnaire layer is closest to this family.

Round-trip and self-consistency. Amrollahi et al. propose verbalising a formal statement back to natural language and checking consistency. Our round-trip baseline achieves AUROC 0.710 on NL-FOL, below the LLM judge.

Cross-check consistency. SAC3 detects hallucinations in black-box LLMs by sampling diverse perturbations and checking consistency. Our cross-system consensus uses a similar principle with z3 equivalence rather than textual similarity.

Automated alignment for autoformalization. FormalAlign trains a model on autoformalization and alignment scoring via a dual contrastive-generative loss. Our approach uses no learned alignment model.

Cross-model translation confidence. Bayless et al. apply the same score form as `c_score` to NL-to-SMT translation with a fixed schema. NoTB uses four LLM families for requirements-to-temporal-logic translation.

N-version programming. The consensus mechanism is an instance of N-version programming. Chen and Avizienis noted that “faulty but identical results may outvote correct results”—precisely the endorsement failure mode we measure ($e = 0.124$ on E, 0.459 under CSC).